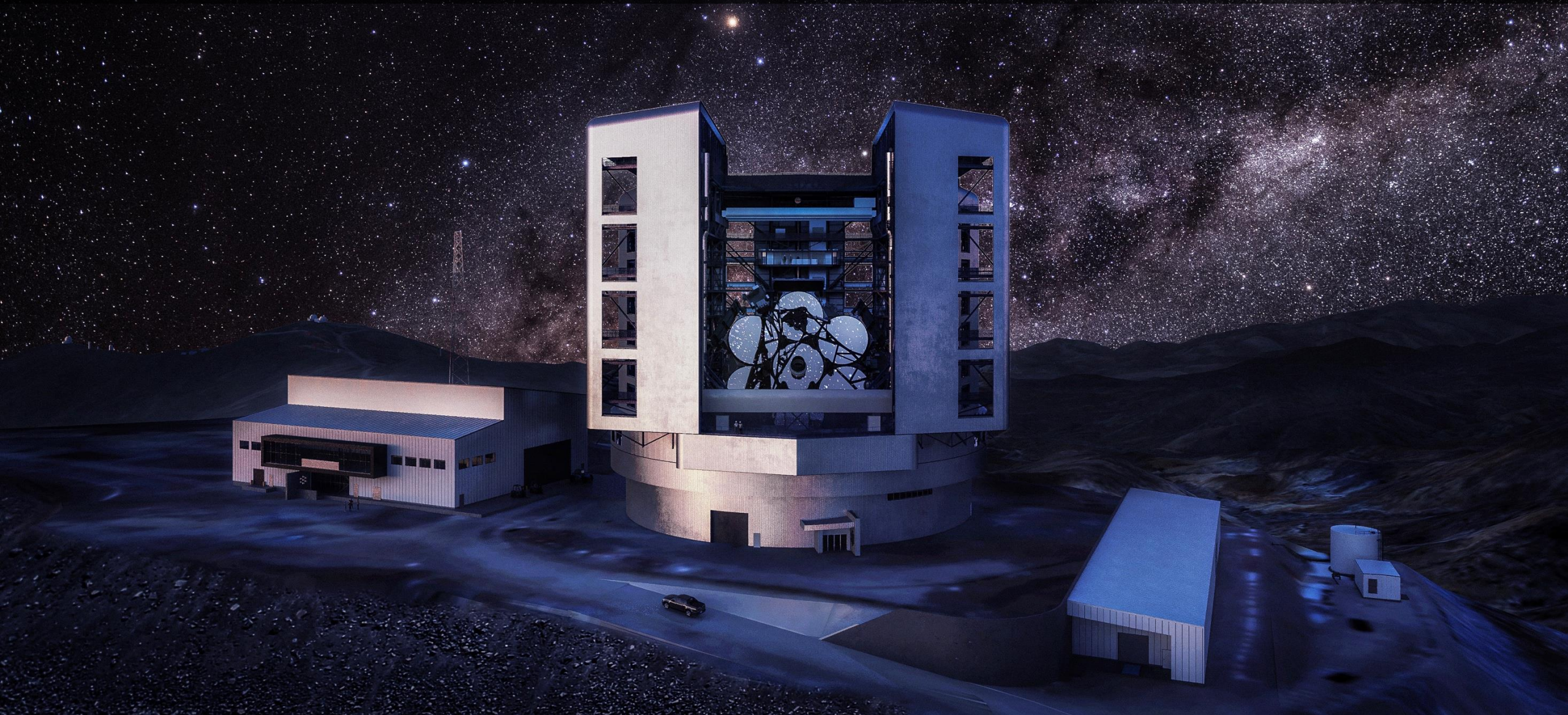


ASTR 302 Fall 2026

Optical Telescopes



Agenda – Sept 23, 2026

- Undergrad TAC Homework due Sunday September 27th 11:59pm
- Optical Telescopes
- Photometry
- This class is going off the rails in OCTOBER SORRY!
 - Y'ALL START OBSERVING
 - Dr. Miles going to ASU, October 7th
 - Dr. Miles MMT Run, October 26 – 27th

Recommended Reading: Birney Chapter 6
Y'all need to read to learn about telescope mounts

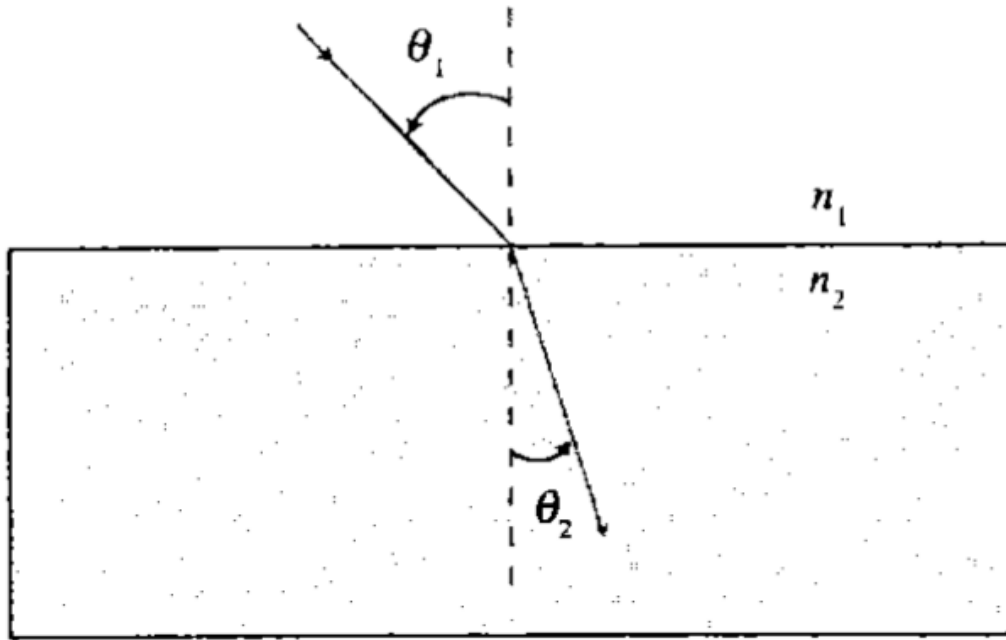
What is the Point of a Telescope?????



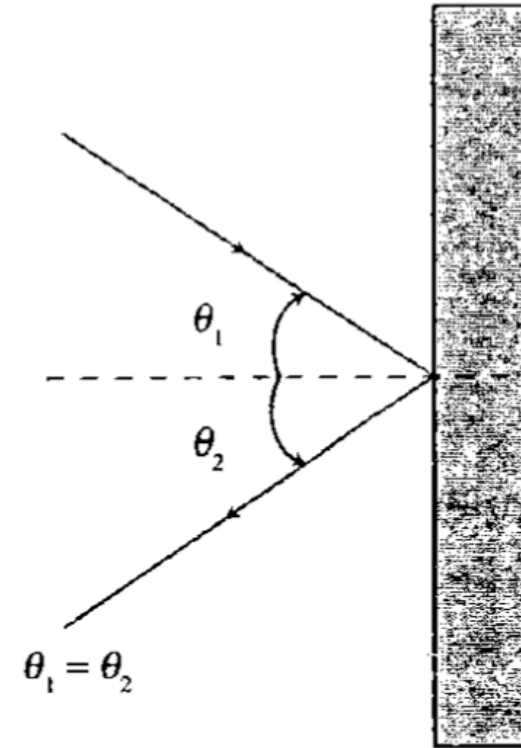
Gemini North Observatory on Maunakea

- To collect significantly more light than what your eye can.
- Magnify far away objects.
- To study objects in wavelengths your eyes cannot see.
- To answer big questions in astronomy
- **(Your answer here)** Inspire wonder

Telescopes either Refract and/or Reflect Light

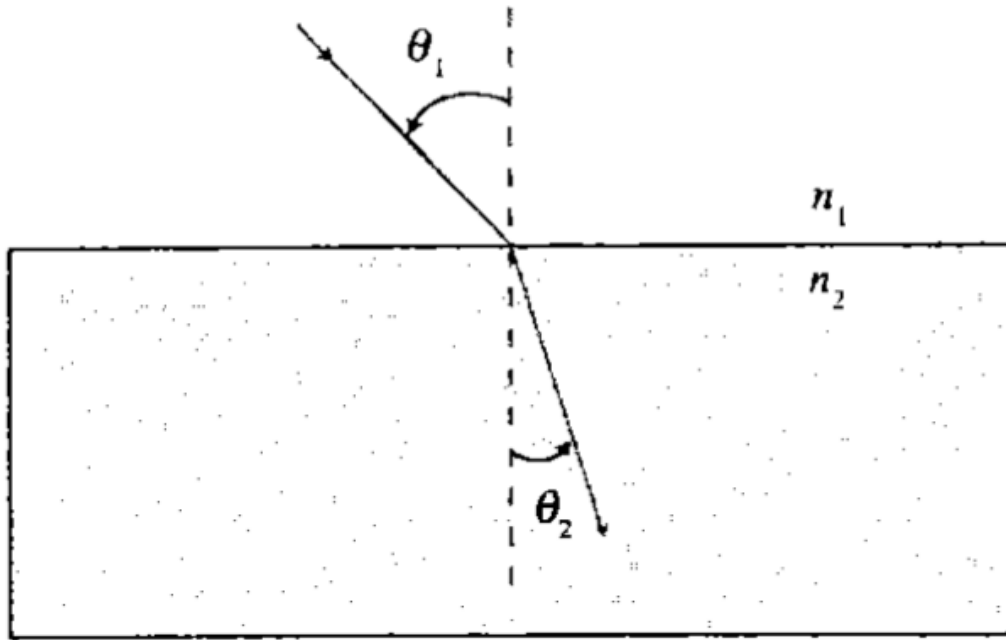


Refraction – is the bending of a wave through a transparent medium.



Reflection – when light is redirected at the boundary of two different mediums back into the medium it originated.

Refraction



The **Law of Refraction** or Snell's Law describes how much light is bent or refracted within a medium.

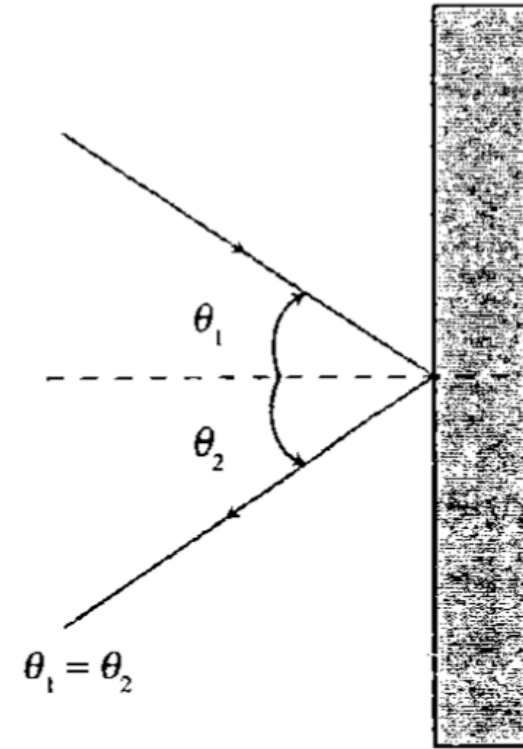
The index of refraction (n) is determined by the speed of light (c) compared the lights speed within a medium (v).

$$n_1 * \sin(\theta_1) = n_2 * \sin(\theta_2)$$

Refraction – is the bending of a wave through a transparent medium.

Reflection

When light reflects the angle of incidence (θ_1) always equals the angle of reflection (θ_2).



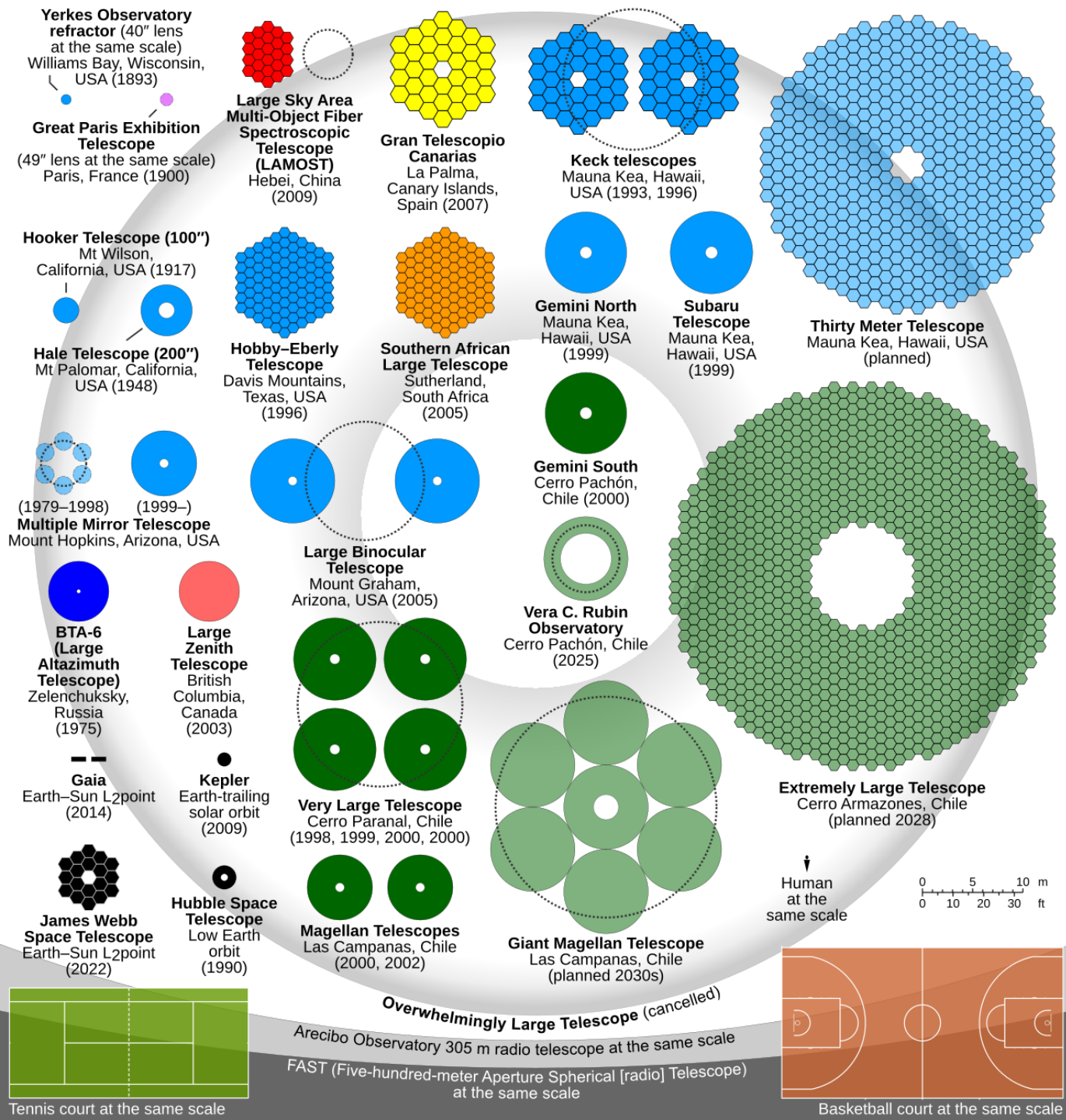
Reflection – when light is redirected at the boundary of two different mediums back into the medium it originated.

The Objective Lens / Mirror

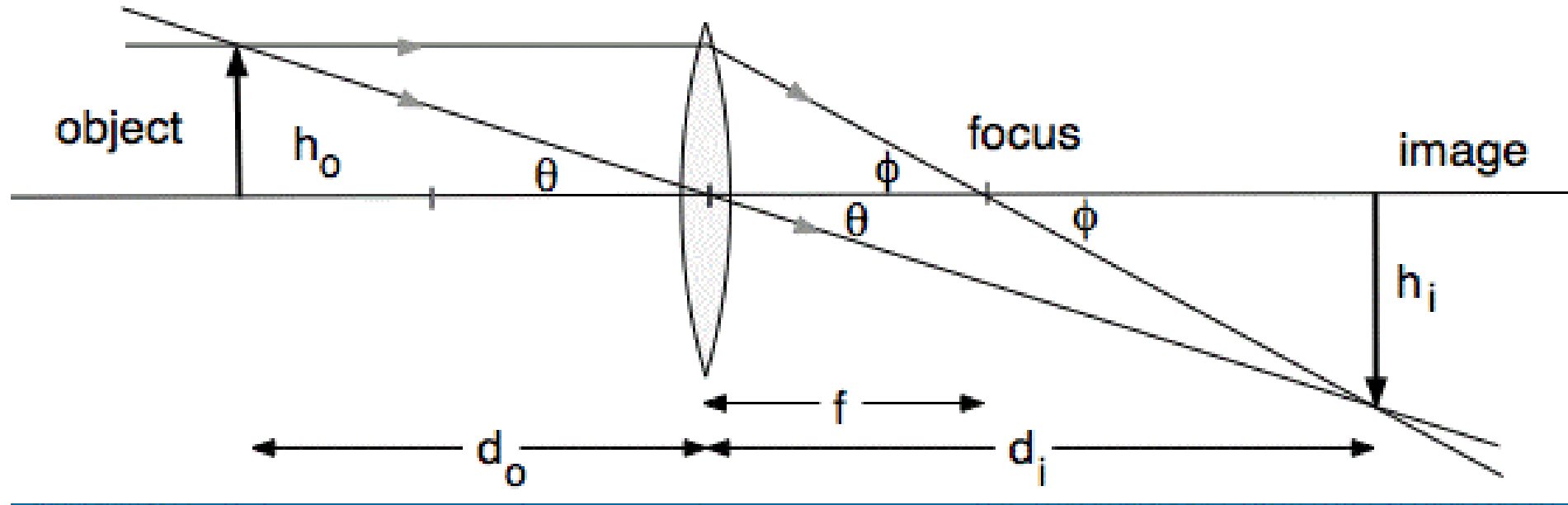
The objective lens/mirror determines the collecting power of the telescope. Collecting power scales with telescope area.

Assuming a circular area the scaling relationship is $\sim R^2$.

Fabricating large lenses and mirrors is challenging because of their weight. **Many large telescopes are segmented.**



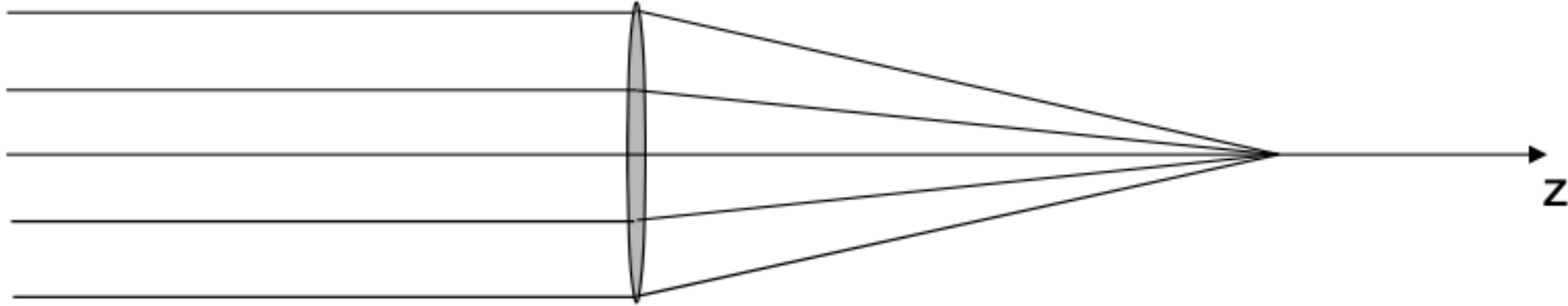
Lens Equations and Properties



The **focal length (f)** of a lens is where all parallel light comes to a focus. The thin lens equation describes how objects will be reimaged.

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$$

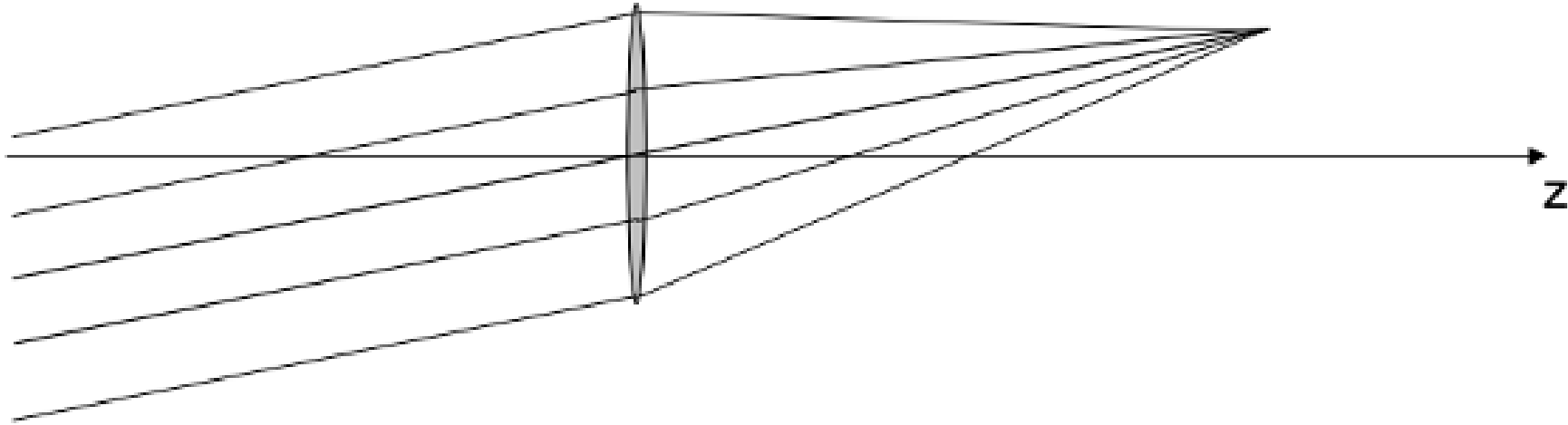
Lens Equations and Properties



If an object is at infinity, it is reimaged at the focal length of the primary lens/mirror. This applies to telescopes looking at objects on the celestial sphere.

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f}$$

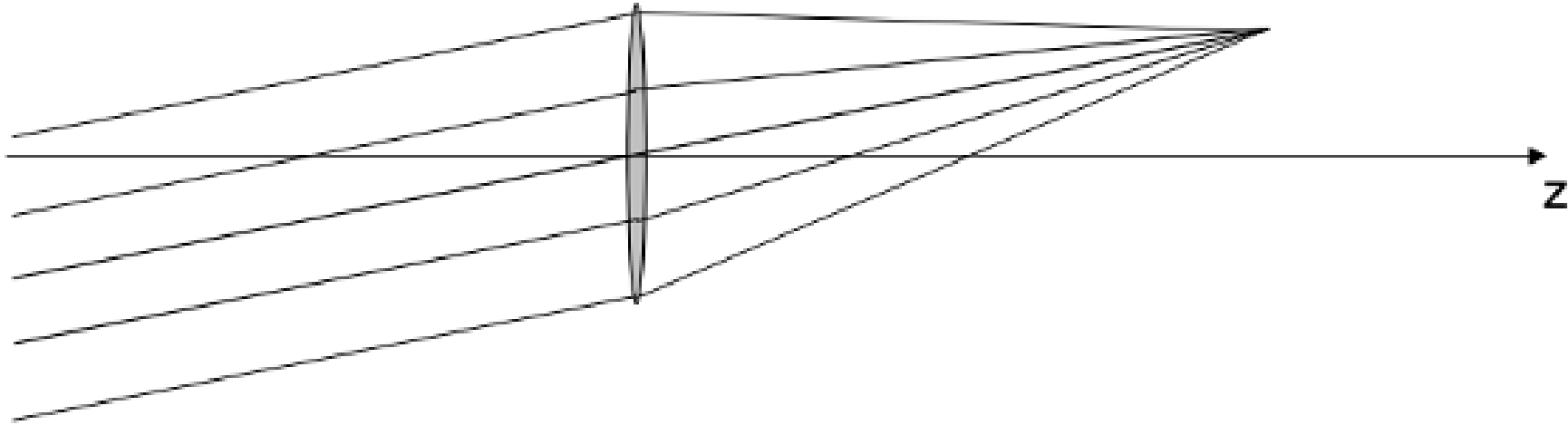
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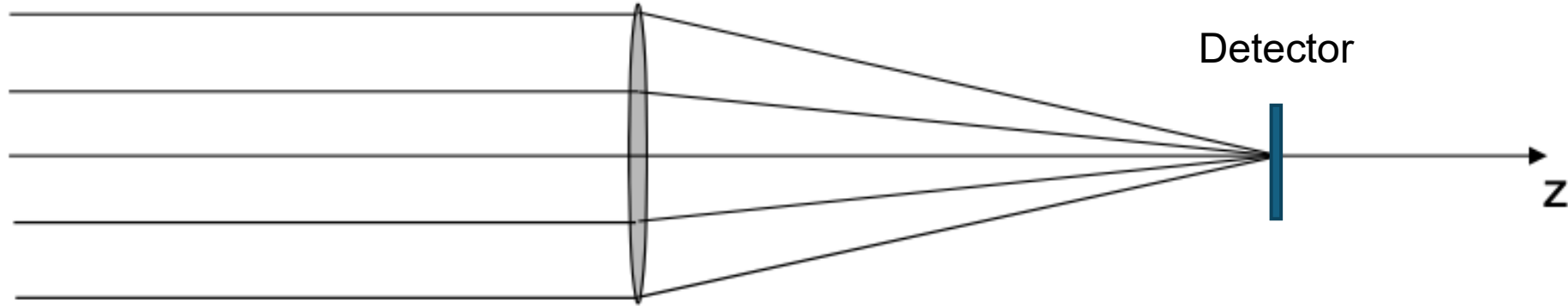
Lens Equations and Properties



The **f#** or focal ratio is the ratio of a telescope's focal length to diameter.

$$f \# = \frac{f}{D}$$

Image Scale of CCD Behind a Lens

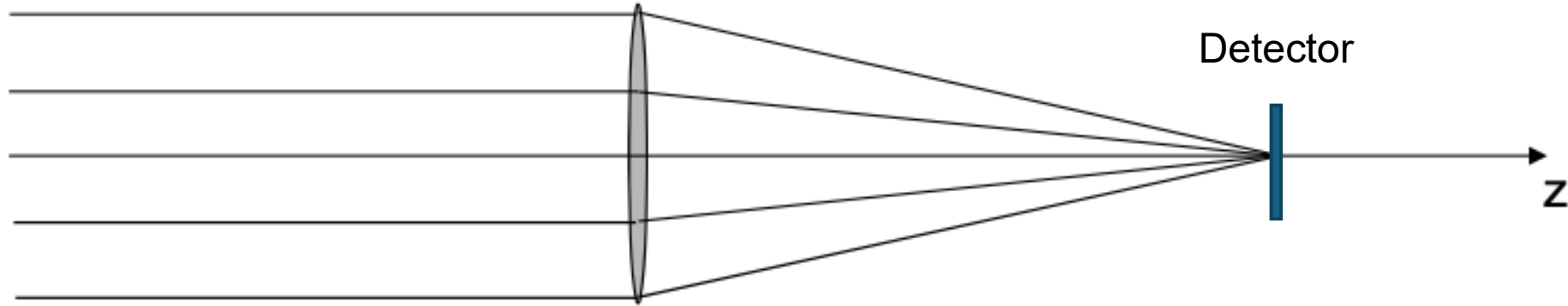


If you know the $f\#$ and the diameter of the telescope you can predict the **plate scale** or **image scale** of a detector with a pixel size (μ). The plate scale tells you how much sky (ex. arcseconds) is represented in each pixel.

$$\text{Plate Scale (arcseconds/pixel)} = \frac{206265 * \mu}{D * f\#}$$

Image Scale of CCD Behind a Lens:

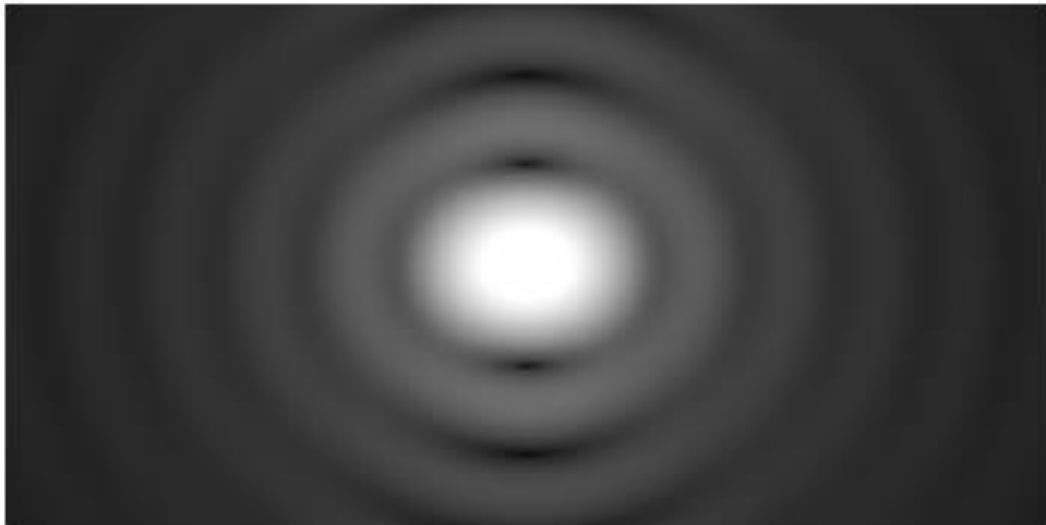
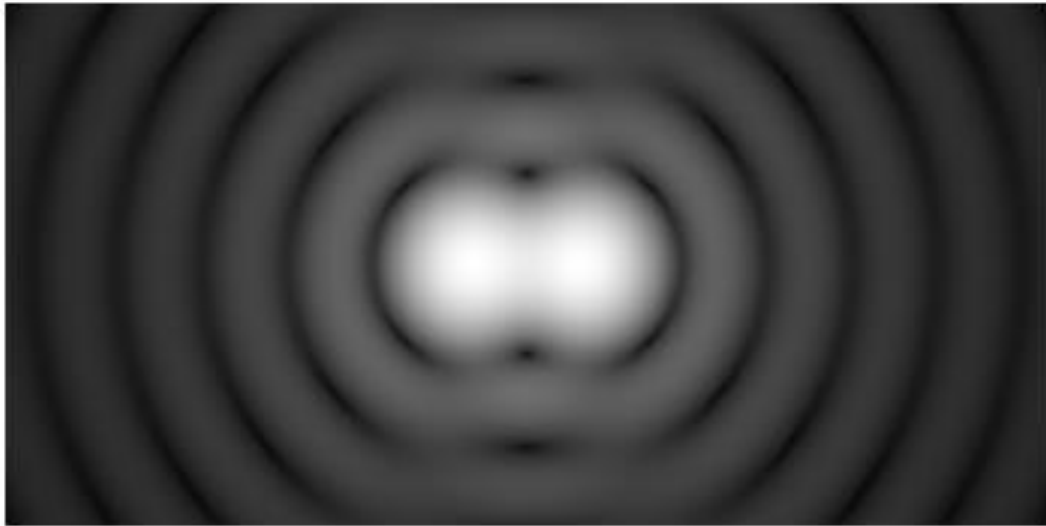
Example 90 inch Prime Camera



If you know the $f\#$ and the diameter of the telescope you can predict the **plate scale** or **image scale** of a detector with a pixel size (μ). The plate scale tells you how much sky (ex. arcseconds) is represented in each pixel.

$$\text{Plate Scale (arcseconds/pixel)} = \frac{206265 * \mu}{D * f\#}$$

The Diffraction Limit



Two Airy Ring Patterns

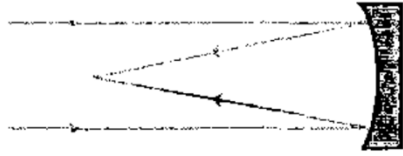
When light propagates through a telescope with a circular primary it appears as an Airy disk/ring due to diffraction. This is due to light's wave behavior.

The **diffraction limit** or the Rayleigh criterion sets physical limit of possible spatial resolution for a telescope at a given wavelength.

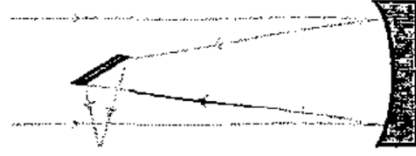
$$\theta \sim 1.22 \frac{\lambda}{D}$$

Without active corrections, most telescopes do not work at this limit.

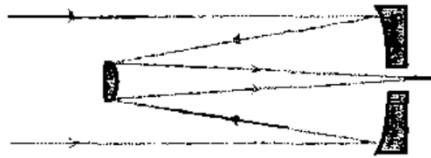
Telescope Designs



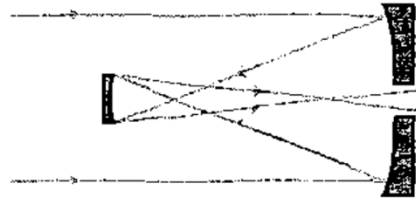
Prime focus



Newtonian



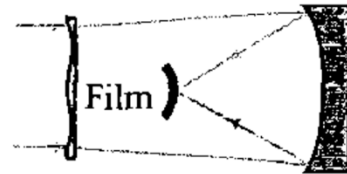
Cassegrain



Gregorian



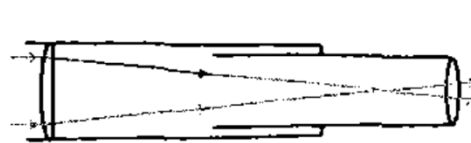
Schmidt-Cassegrain



Schmidt Camera



Maksutov



Refractor

There are several types of telescope designs. You can add fold mirrors or other optics to decrease the physical size of your telescope.

Larger telescopes tend to have a Cassegrain/Gregorian approach.

Telescope Mounts



Telescope mounts have three jobs:

- Support the telescope
- Point to relevant targets
- Track targets

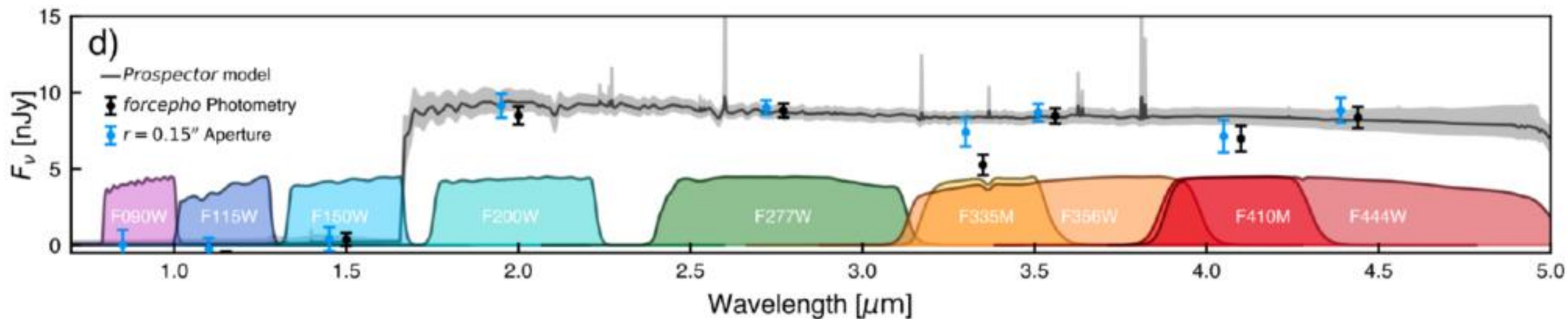
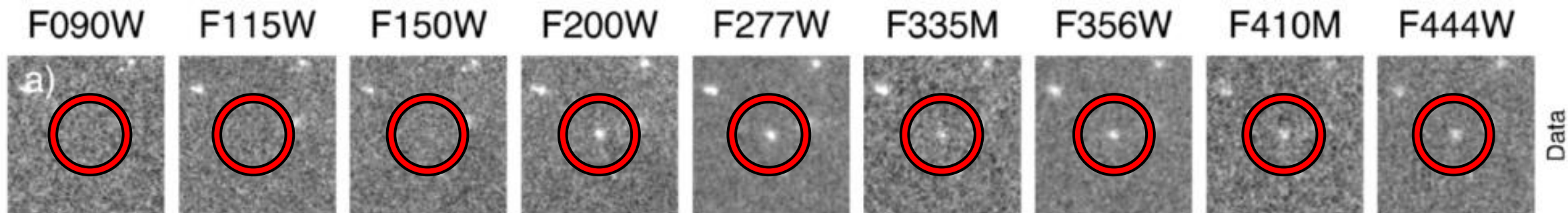
There two main types of telescope mounts:

- Equatorial Mounts
- Altitude-Azimuthal Mounts



ASTR 302 Fall 2026: Photometry

JADES-GS-z12-0, $z = 12.63$

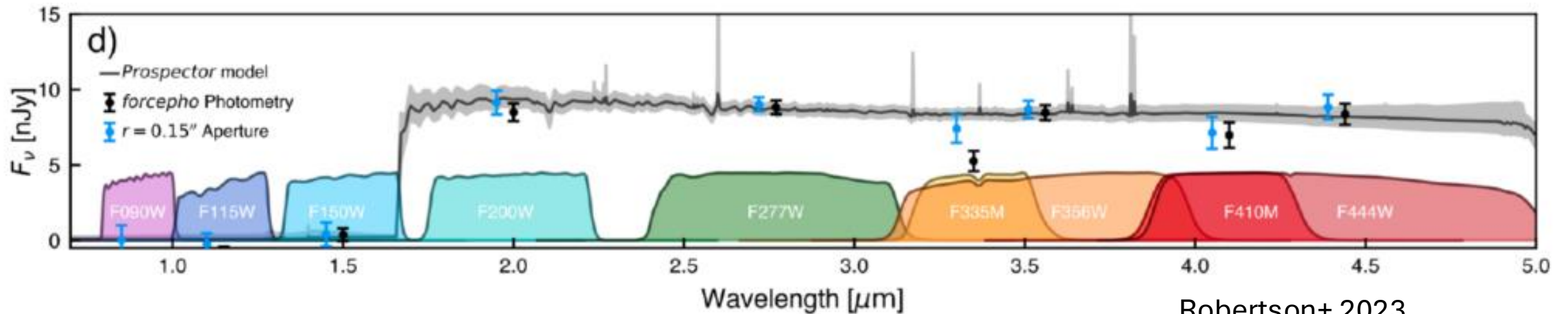
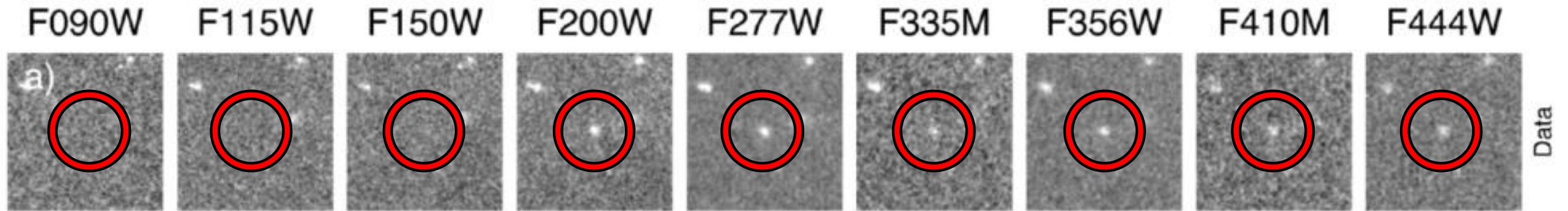


Robertson+ 2023

(<https://ui.adsabs.harvard.edu/abs/2023NatAs...7..611R/abstract>)

Photometry

JADES-GS-z12-0, $z = 12.63$

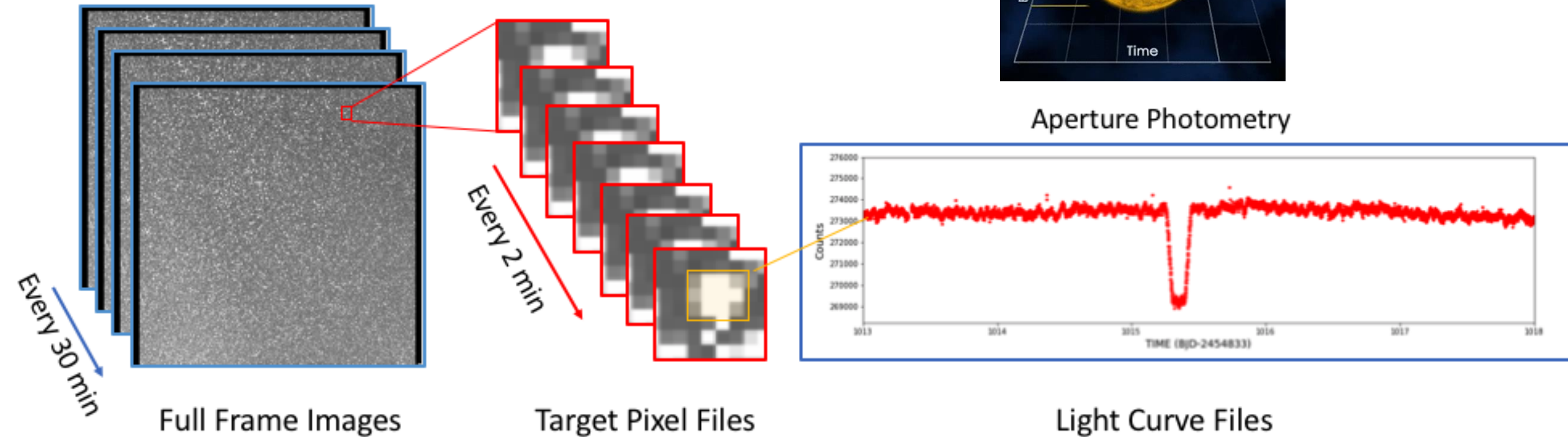


Robertson+ 2023

(<https://ui.adsabs.harvard.edu/abs/2023NatAs...7..611R/abstract>)

We use **photometry** to determine how bright things are.

Photometry



TESS Data Archive Figure

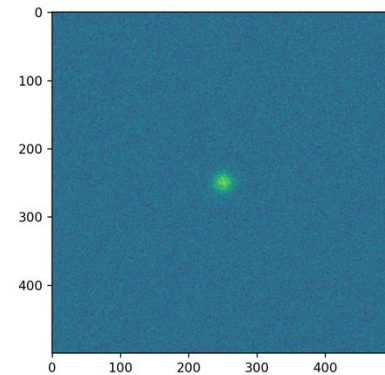
- We use **relative photometry** to determine:
- how bright targets are compared to other objects
 - how the brightness of an object changes over time

Photometry: Important Concepts to Understand

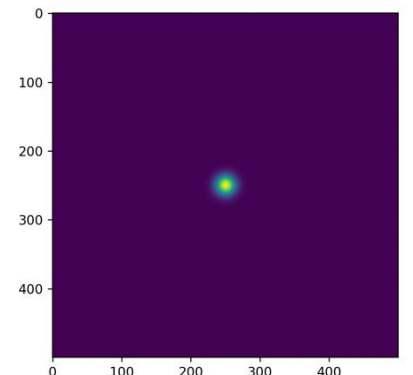
If your detector/CCD has a linear response to light, the amount of counts you measure for a fixed exposure time should be linear with target brightness.

$$m - m_{ref} = -2.5 \log_{10} \left(\frac{F}{F_{ref}} \right)$$

Science Target



Reference Target



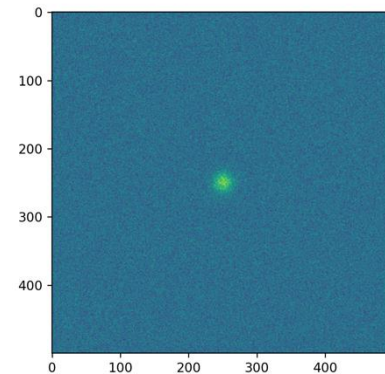
Photometry: Important Concepts to Understand

If your detector/CCD has a linear response to light, the amount of counts you measure for a fixed exposure time should be linear with target brightness.

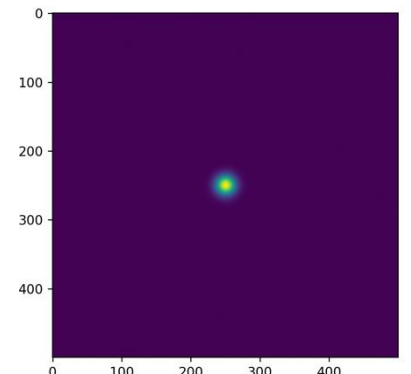
$$m - m_{ref} = -2.5 \log_{10} \left(\frac{F}{F_{ref}} \right)$$

$$m - m_{ref} = -2.5 \log_{10} \left(\frac{C}{C_{ref}} \right)$$

Science Target



Reference Target



Photometry: Important Concepts to Understand

Most filters have a **magnitude zero point**. There is an absolute flux value associated with that zero point.

$$m - m_0 = -2.5 \log_{10} \left(\frac{F}{F_0} \right)$$

$$m - m_0 = -2.5 \log_{10} \left(\frac{C}{C_0} \right)$$

SVO Filter Service:

<https://ui.adsabs.harvard.edu/abs/2020sea..confE.182R/abstract>

Photometry: Important Concepts to Understand

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Photometry: Important Concepts to Understand

Most filters have a **magnitude zero point**. There is an absolute flux value associated with that zero point. The **instrumental zero point** is where you have 1 count/s response.

$$m - m_0 = -2.5 \log_{10} \left(\frac{F}{F_0} \right)$$

$$m - m_0 = -2.5 \log_{10} \left(\frac{C}{C_0} \right)$$

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Photometry: Important Concepts to Understand

Most filters have a **magnitude zero point**. There is an absolute flux value associated with that zero point. The **instrumental zero point** is where you have 1 count/s response.

$$m - m_0 = -2.5 \log_{10} \left(\frac{F}{F_0} \right)$$

$$m - \boxed{m_0} = -2.5 \log_{10} \left(\frac{C}{\boxed{C_0}} \right) \quad C_0 = 1 \text{ count/s}$$

SVO Filter Service:

<https://ui.adsabs.harvard.edu/abs/2020sea..confE.182R/abstract>

Photometric Standard Stars

- Standard stars are stars with precisely measured magnitudes (better than ± 0.01 mag)
- Standard stars should have the following qualities:
 - Observed at similar airmass/atmospheric conditions as the science target(s).
 - Should not be variable.
 - Should not be secret binary star (spectroscopic binary).
 - Ideally be brighter than your science observations.

Finding Photometric Standard Stars: Large Fields

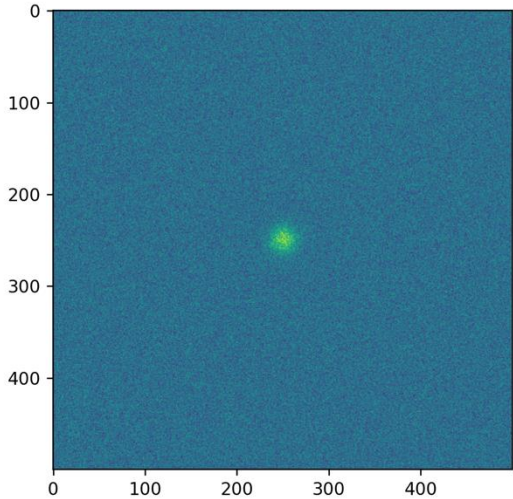


Photo from Peter Milne

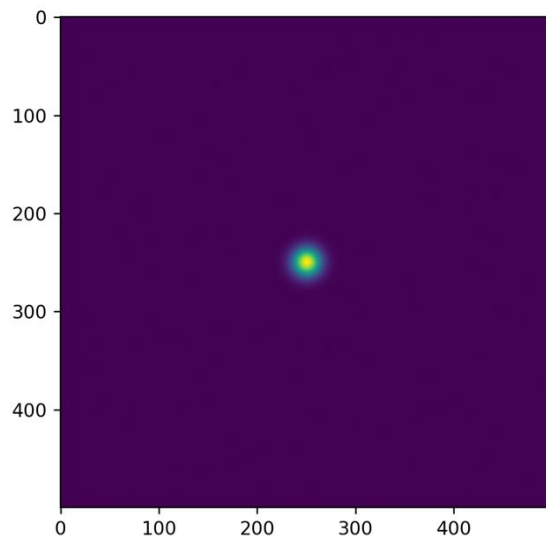
- If your CCD/detector images cover enough sky you can check if any of the point sources are standard stars.
- The Kuiper 61" website has finder charts with Landolt+ 1992 standard stars for the entire year. (RA: 0 –24 hours)
- If there are multiple stars in the field of view, you can get a better absolute photometric calibration.
- Ignoring the impact of extinction, etc. The atmospheric conditions should be similar across the frame.

Finding Photometric Standard Stars

Science Target

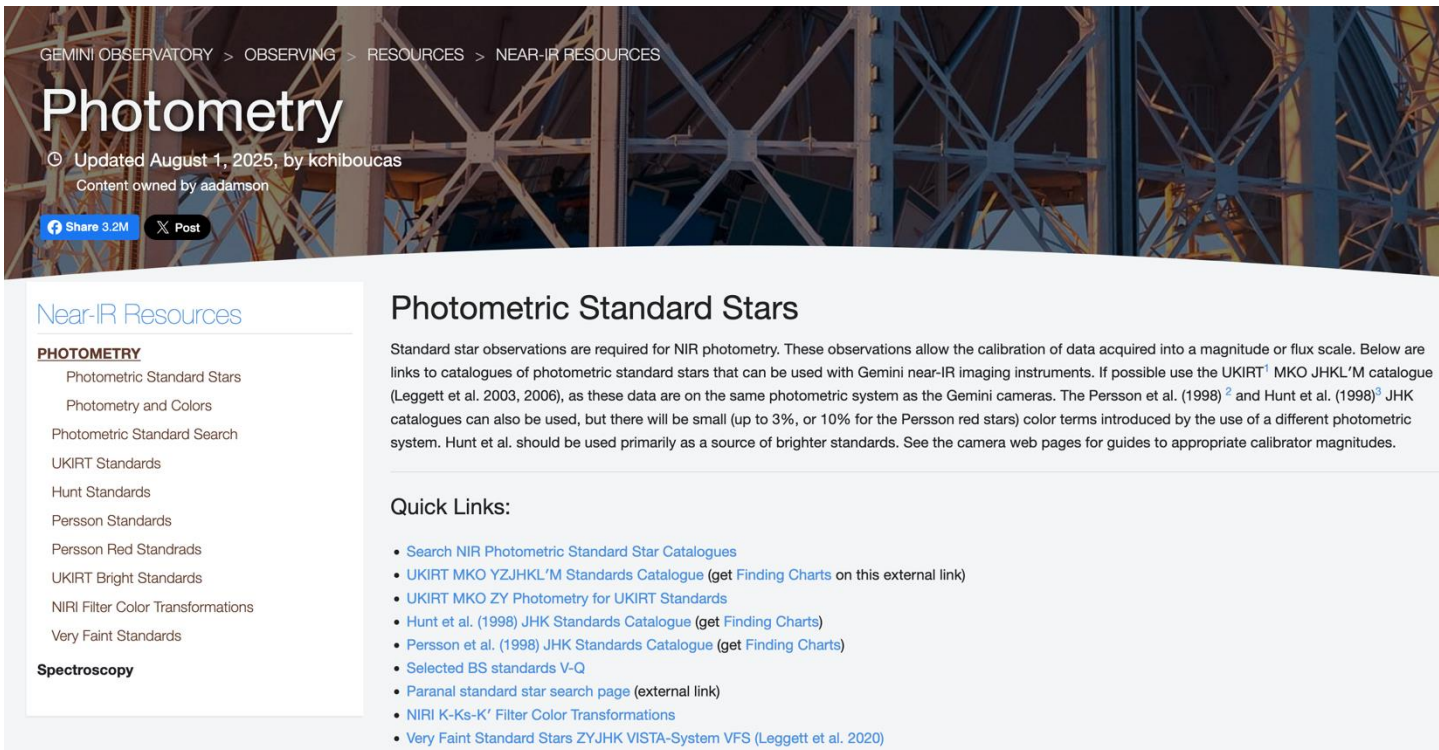


Reference Target



- If your CCD/detector images are small, you may need to point at a separate standard at some point over the night.
- Sometimes your field may not have a suitable standard and you still need dedicated photometric calibrations.
- Your standard stars must be observed at similar conditions (airmass, etc) as your science.

Finding Photometric Standard Stars

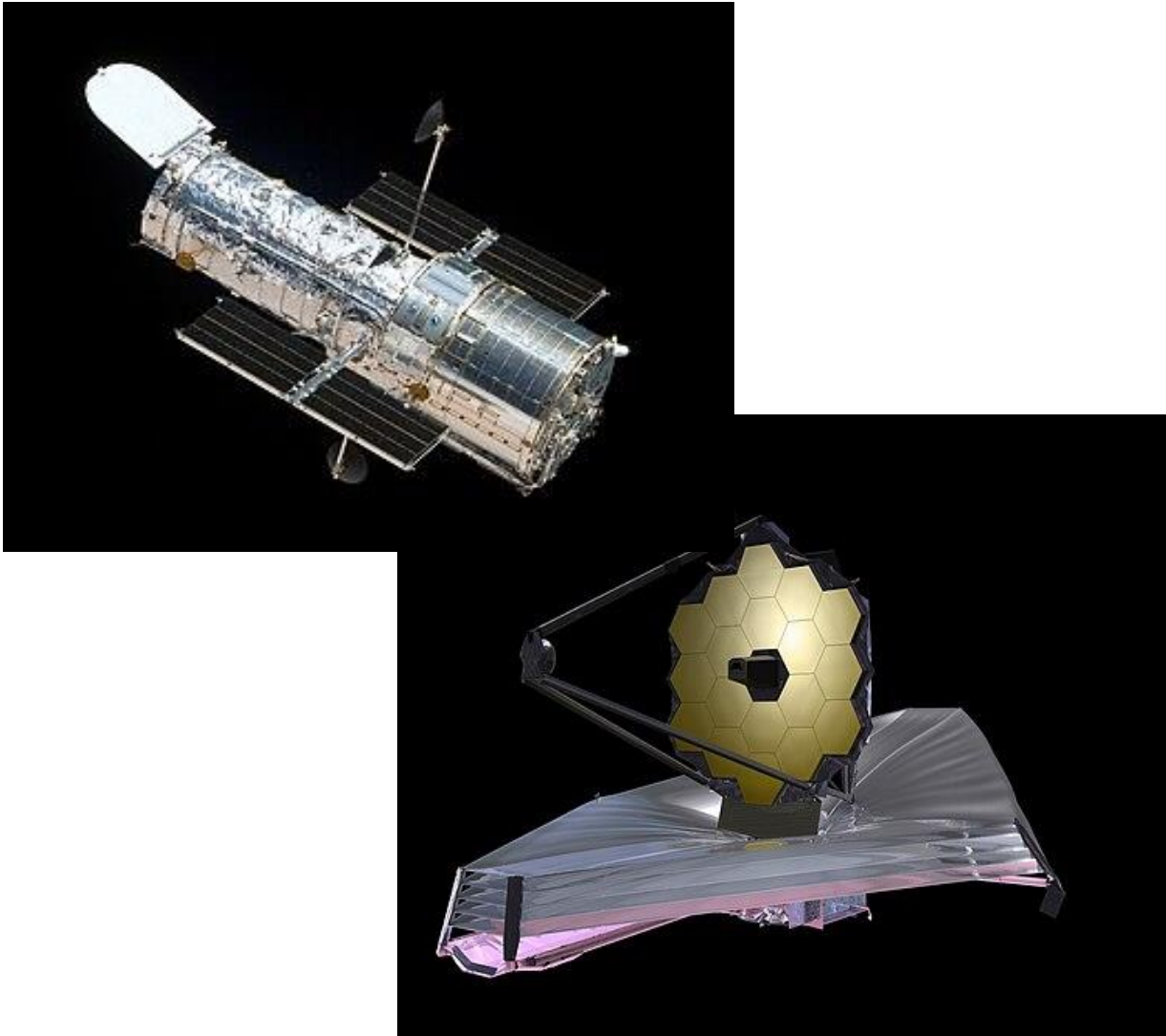


The screenshot shows the Gemini Observatory website's 'Photometry' page under the 'Near-IR Resources' section. The page has a header with navigation links: GEMINI OBSERVATORY > OBSERVING > RESOURCES > NEAR-IR RESOURCES. The main title 'Photometry' is prominently displayed, with a sub-header indicating it was updated on August 1, 2025, by kchiboucas, and content is owned by aadamson. Social sharing buttons for Facebook (3.2M shares) and a 'Post' button are visible. A left sidebar lists various resources under 'Near-IR Resources', including 'PHOTOMETRY' (with sub-links like 'Photometric Standard Stars', 'Photometry and Colors', 'Photometric Standard Search', 'UKIRT Standards', 'Hunt Standards', 'Persson Standards', 'Persson Red Standards', 'UKIRT Bright Standards', 'NIRI Filter Color Transformations', and 'Very Faint Standards'), and 'Spectroscopy'. The main content area is titled 'Photometric Standard Stars' and contains a paragraph explaining the need for standard star observations for NIR photometry, mentioning catalogues like UKIRT¹ MKO JHKL'M, Leggett et al. (2003, 2006), Persson et al. (1998)², and Hunt et al. (1998)³. Below this text is a 'Quick Links' section with a list of links to various catalogues and search pages, including 'Search NIR Photometric Standard Star Catalogues', 'UKIRT MKO YZJHKL'M Standards Catalogue', 'UKIRT MKO ZY Photometry for UKIRT Standards', 'Hunt et al. (1998) JHK Standards Catalogue', 'Persson et al. (1998) JHK Standards Catalogue', 'Selected BS standards V-Q', 'Paranal standard star search page', 'NIRI K-Ks-K' Filter Color Transformations', and 'Very Faint Standard Stars ZYJHK VISTA-System VFS'.

- To start looking for standards I would look at observatory webpages. You can still observe Landolt standards as individual stars if you want.
- Some observatories have advice pages on planning and choosing photometric calibrators (Gemini).

<https://www.gemini.edu/observing/resources/near-ir-resources/photometry>

Finding Photometric Standard Stars: Space



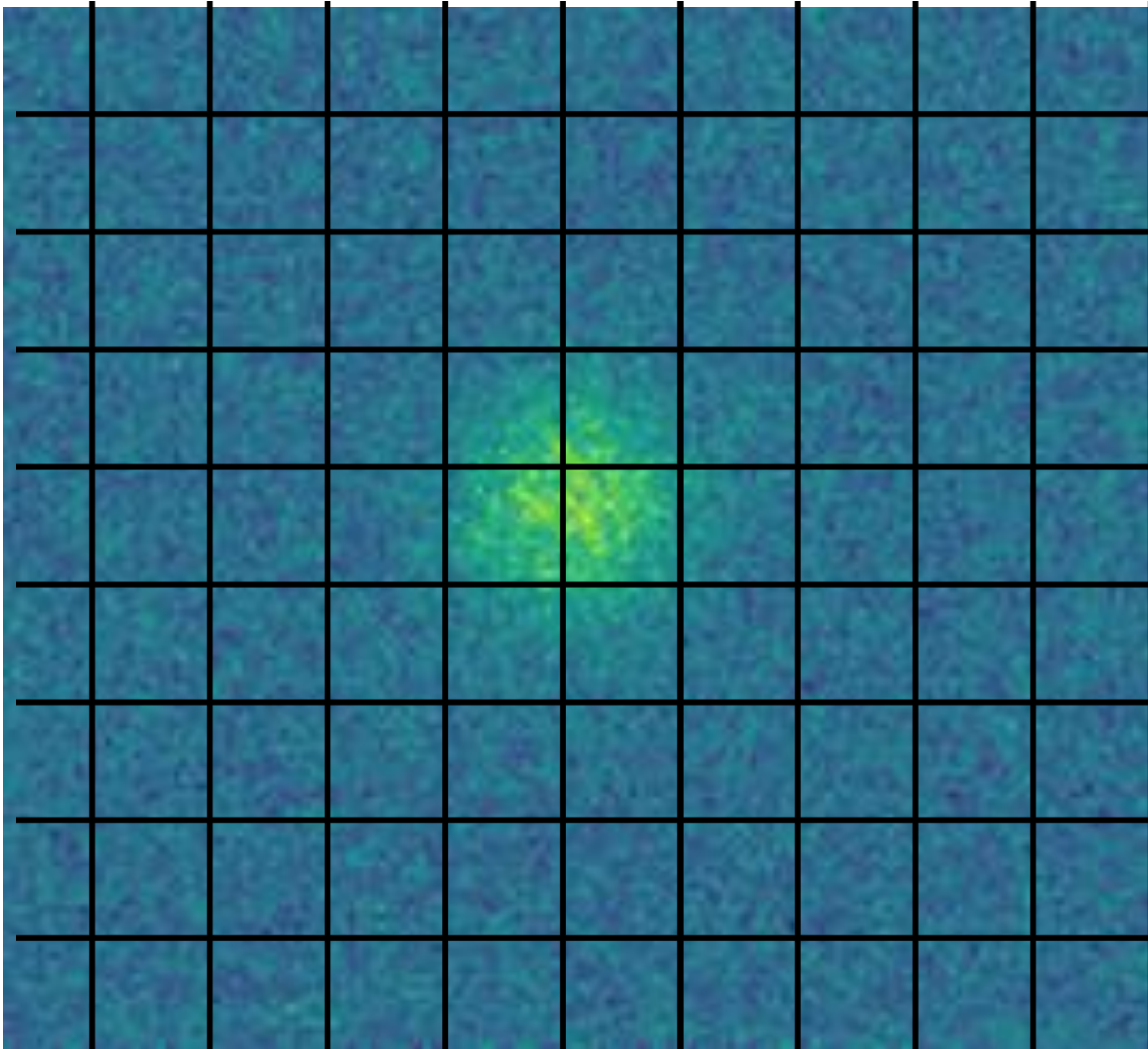
- Space-based telescopes are so sensitive that some standards you would use on the ground could saturate their detectors.
- Space-based observatories often observe white dwarfs as standards because they behave like a black body.
- You do not need to take calibrations as often on the ground.
- STSci maintains a repository of standard star data called CALSPEC

CALSPEC website

<https://www.stsci.edu/hst/instrumentation/reference-data-for-calibration-and-tools/astronomical-catalogs/calspec>

Aperture Photometry: Adding up Light

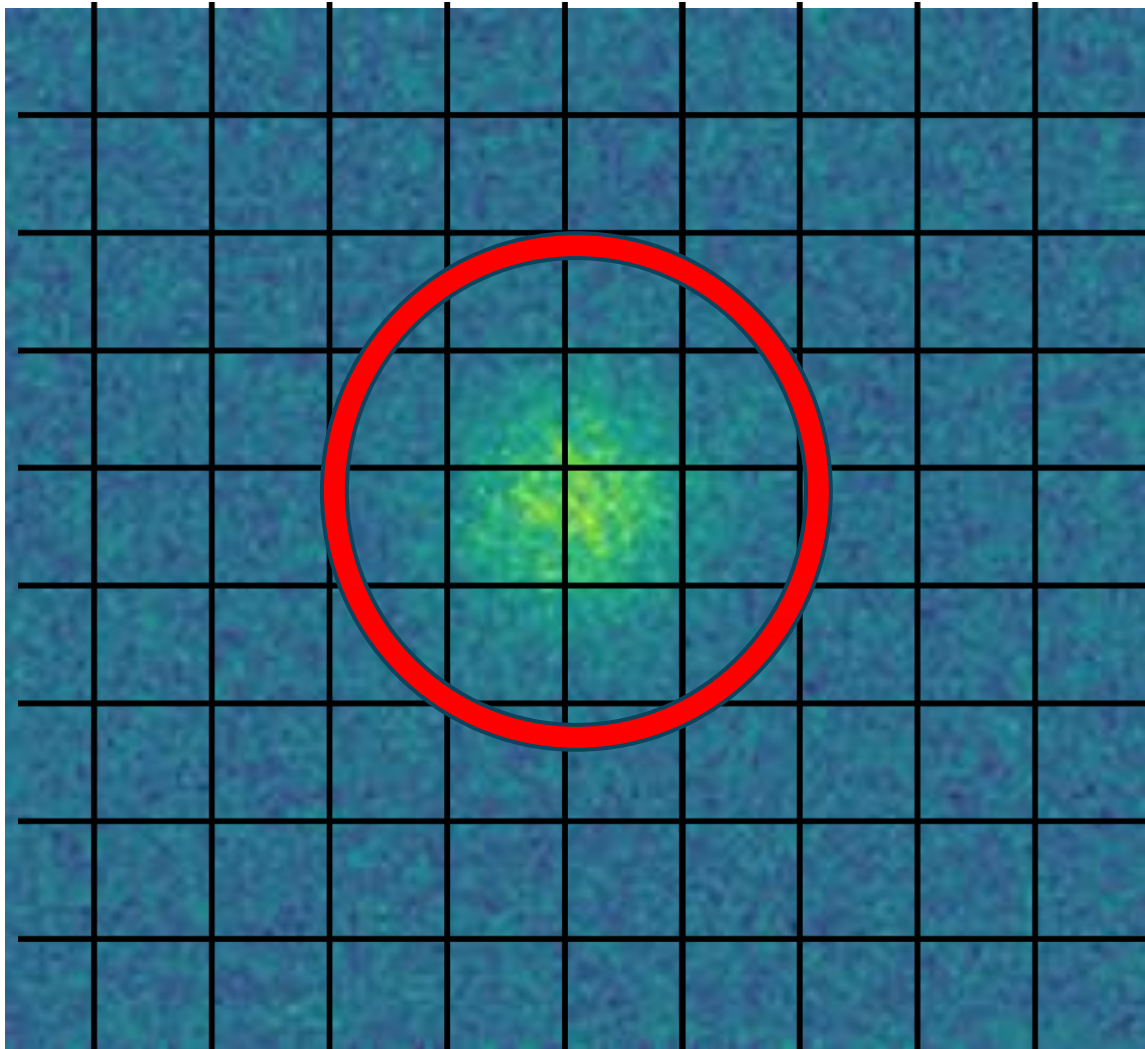
Reduced Image



- To estimate the counts or the response of a target you need to choose an area to add up light over.
- In general we chose a circle because without any optical aberrations the Airy Ring pattern is circular.

Aperture Photometry: Adding up Light

Reduced Image

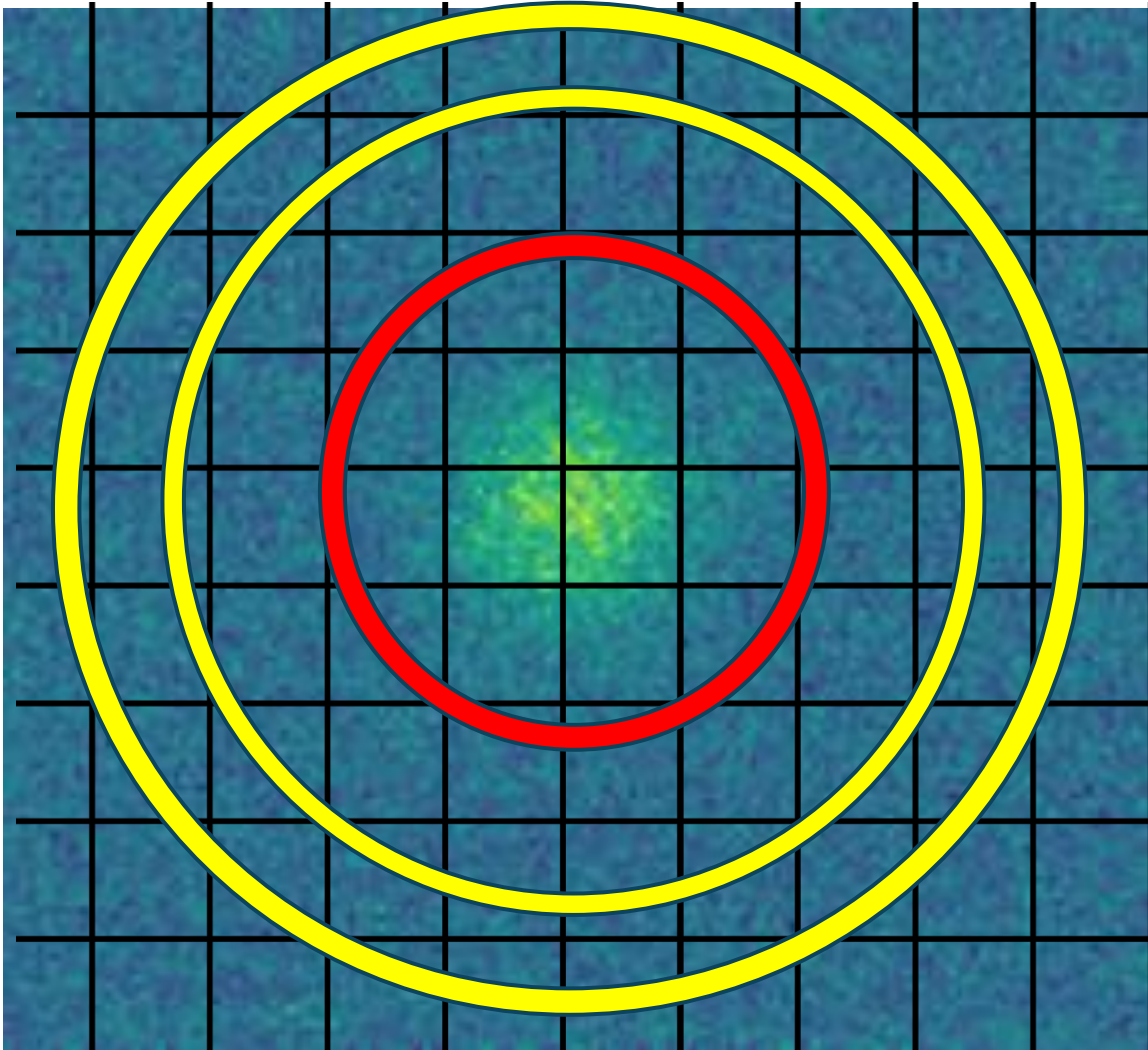


- To estimate the counts or the response of a target you need to choose an area to add up light over.
- In general we chose a circle because without any optical aberrations the Airy Ring pattern is circular.
- You add up the counts within the circle by taking a sum. There are python functions that can do this in astropy.

$$C_{tot} = \sum_{i=1}^{n_{pixels}} C_i$$

Aperture Photometry: Estimating Noise

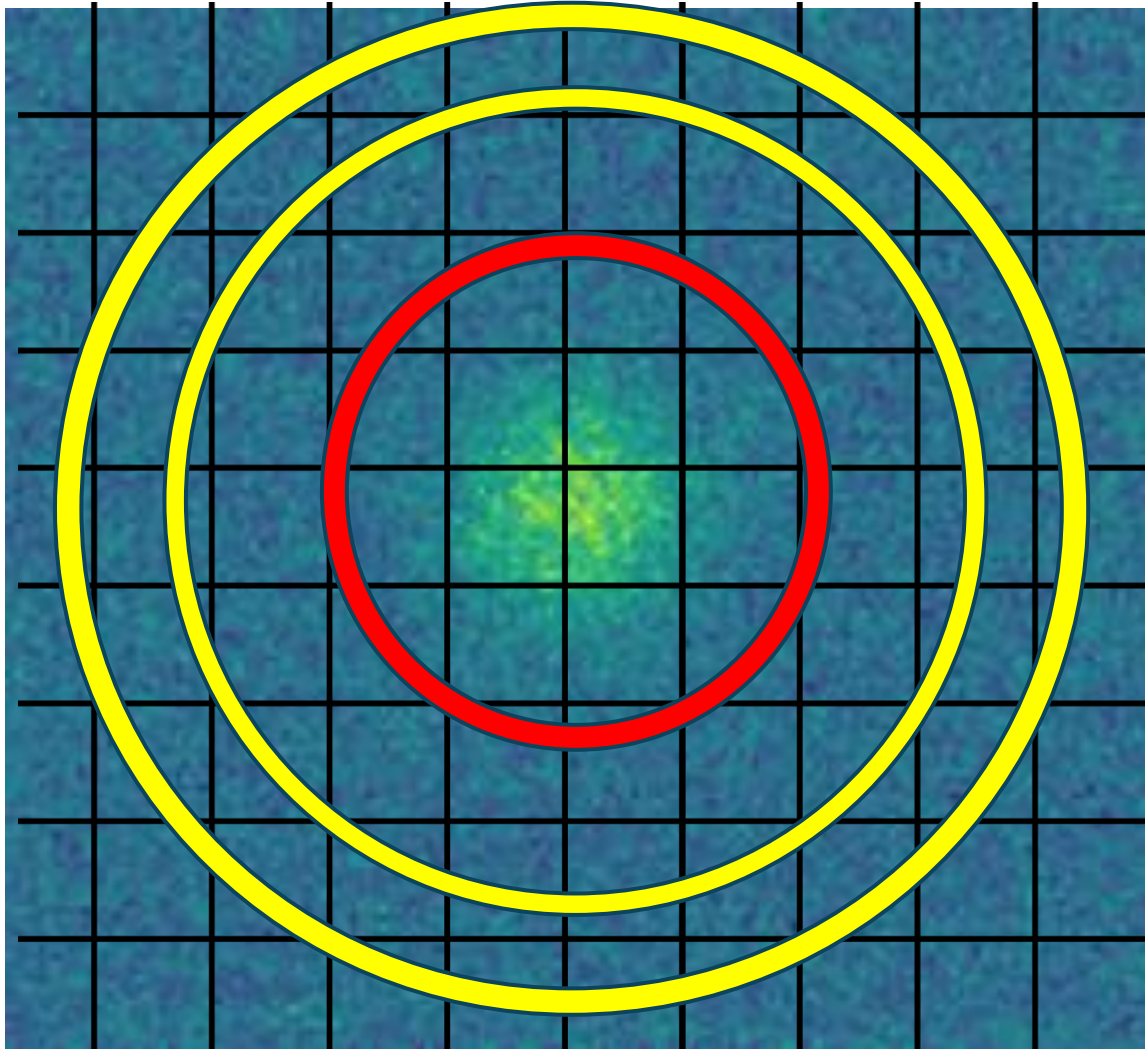
Reduced Image



- You can choose an area away from your science target to estimate the residual random noise from the sky, dark current, etc.
- Many astronomers use an annulus around the science aperture to estimate this.
- Take σ (np.std) of pixels in the annulus.

Aperture Photometry: Estimating Noise

Reduced Image



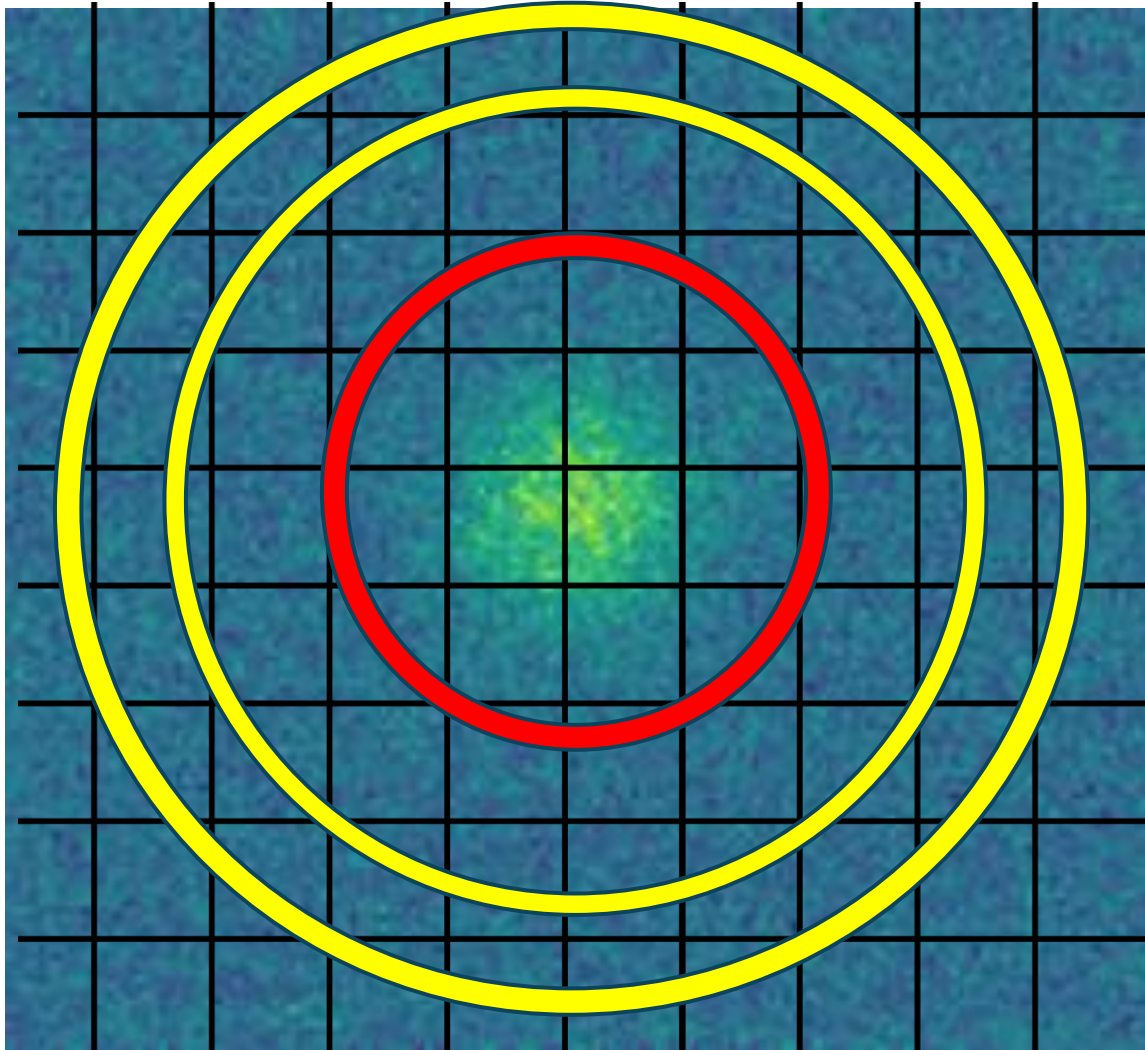
- You can choose an area away from your science target to estimate the residual random noise from the sky, dark current, etc.
- Many astronomers use an annulus around the science aperture to estimate this.

RECALL

$$\sigma_{\text{total}} = \sqrt{\sigma_{\text{star}}^2 + \sigma_{\text{sky}}^2 + \sigma_{\text{telescope}}^2 + \sigma_{\text{detector}}^2}$$

Aperture Photometry: Estimating Noise

Reduced Image



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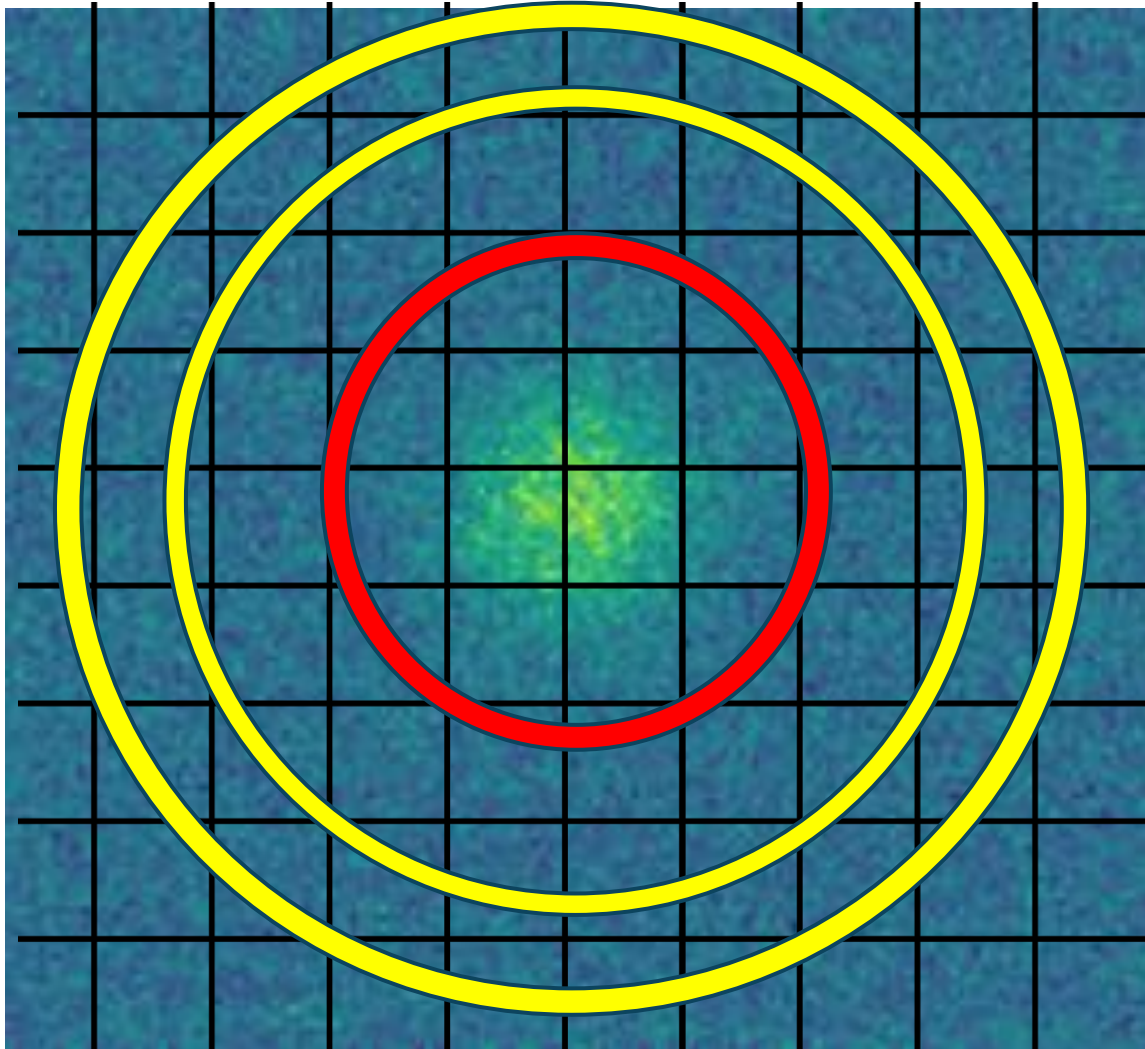
RECALL

$$\sigma_{\text{total}} = \sqrt{\sigma_{\text{star}}^2 + \sigma_{\text{sky}}^2 + \sigma_{\text{telescope}}^2 + \sigma_{\text{detector}}^2}$$

$$\sigma_{\text{total}} = \sqrt{\sigma_{\text{star}}^2 + \sigma_{\text{annulus}}^2}$$

Aperture Photometry: Estimating Noise

Reduced Image



- You can choose an area away from your science target to estimate the residual random noise from the sky, dark current, etc.
- Many astronomers use an annulus around the science aperture to estimate this.

RECALL

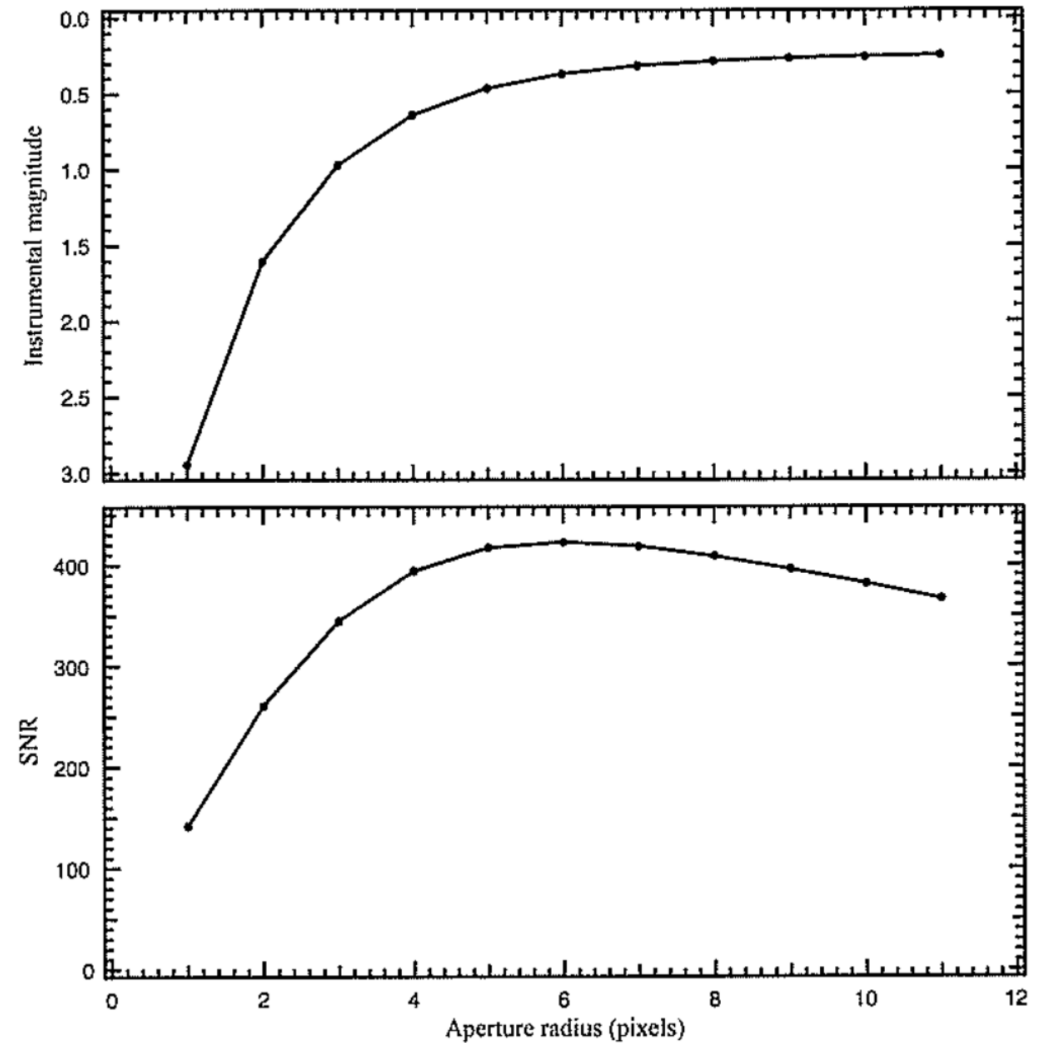
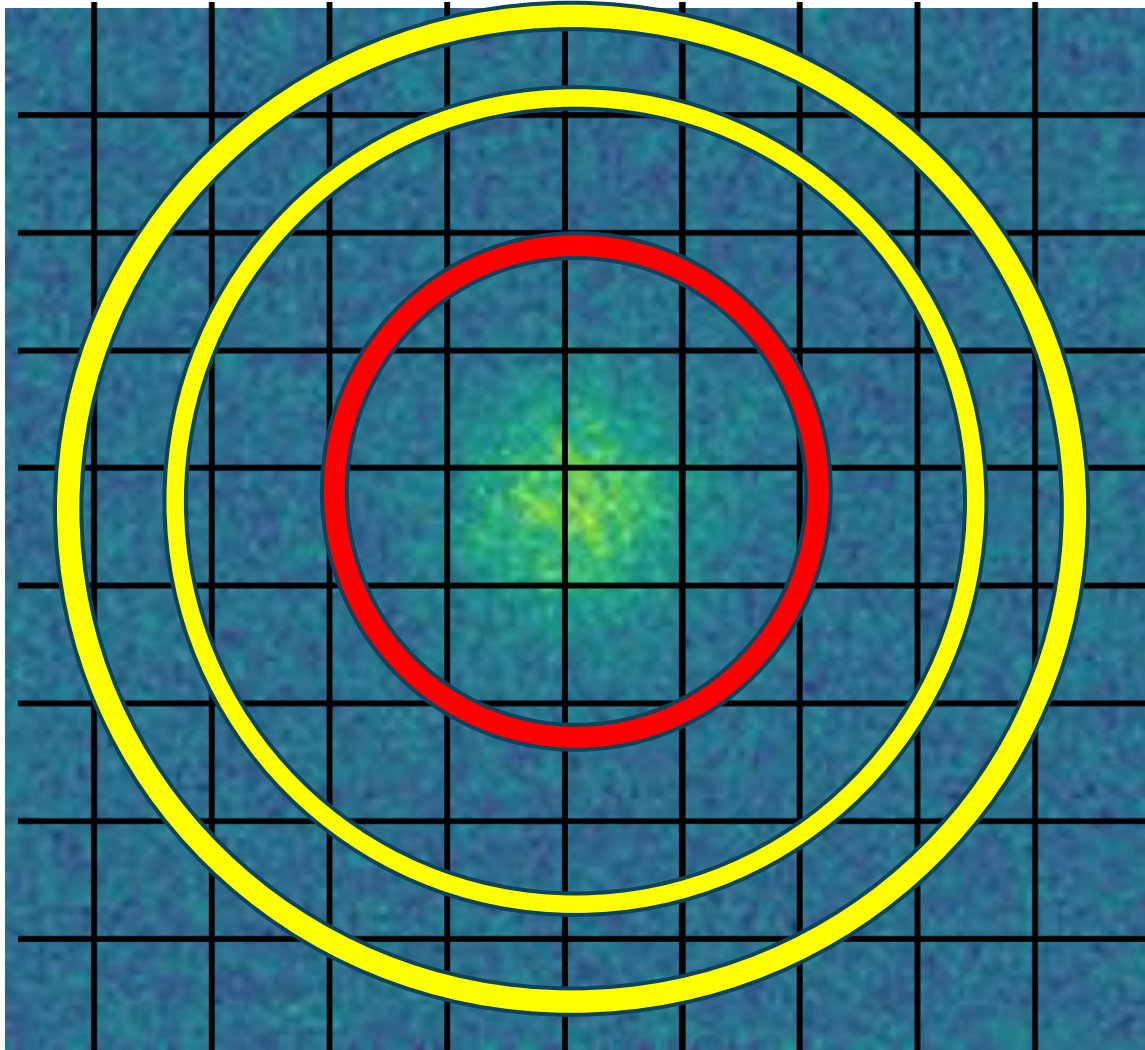
$$\sigma_{\text{total}} = \sqrt{\sigma_{\text{star}}^2 + \sigma_{\text{sky}}^2 + \sigma_{\text{telescope}}^2 + \sigma_{\text{detector}}^2}$$

$$\sigma_{\text{total}} = \sqrt{\sigma_{\text{star}}^2 + \sigma_{\text{annulus}}^2}$$

$$\sigma_{\text{total}} = \sqrt{\text{Counts}_{\text{star}} + \sigma_{\text{annulus}}^2}$$

Aperture Photometry: What Radii do I Pick?

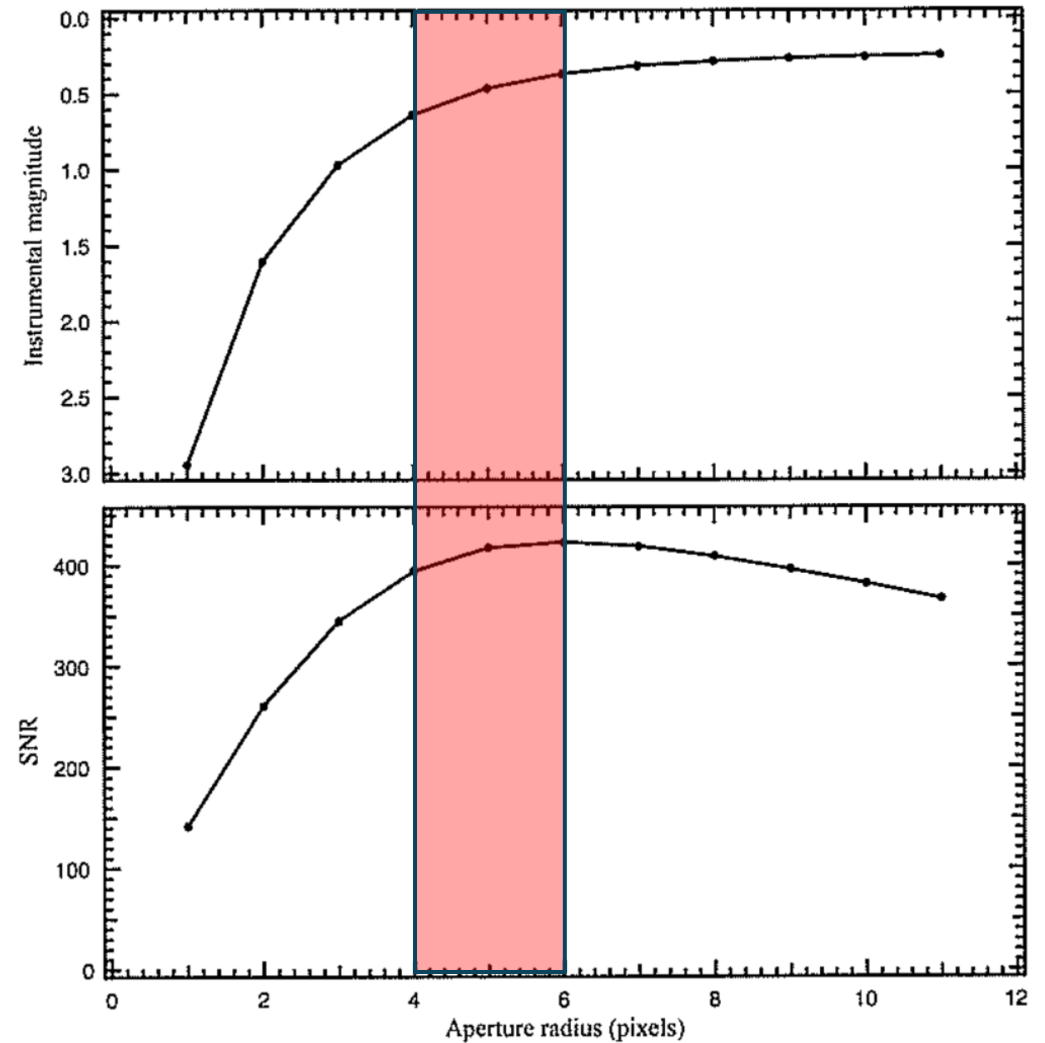
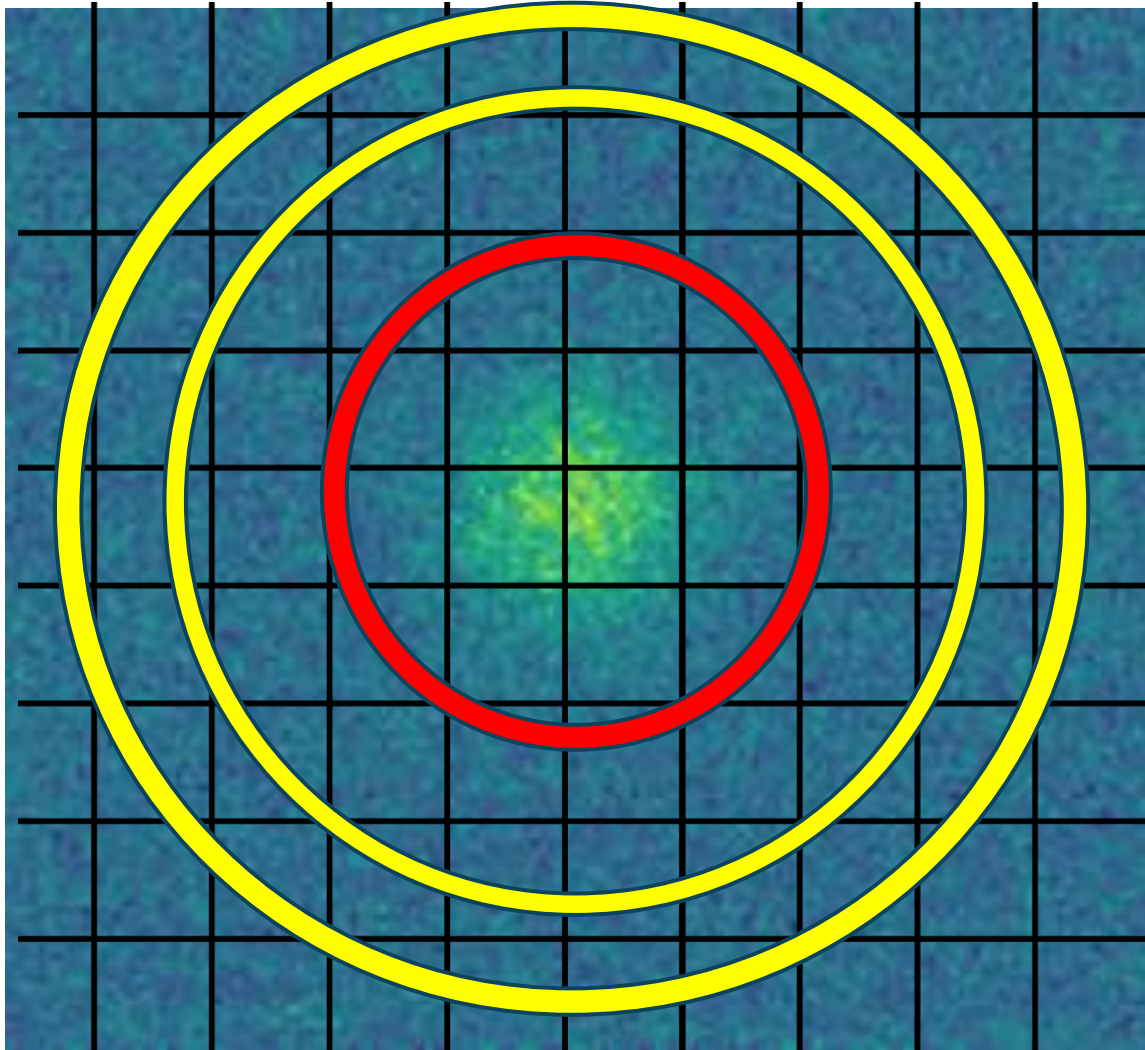
Reduced Image



You usually choose an extraction radius to maximize signal to noise.

Aperture Photometry: What Radii do I Pick?

Reduced Image



You usually choose an extraction radius to maximize signal to noise.

Example: Calibrating Supernova Photometry

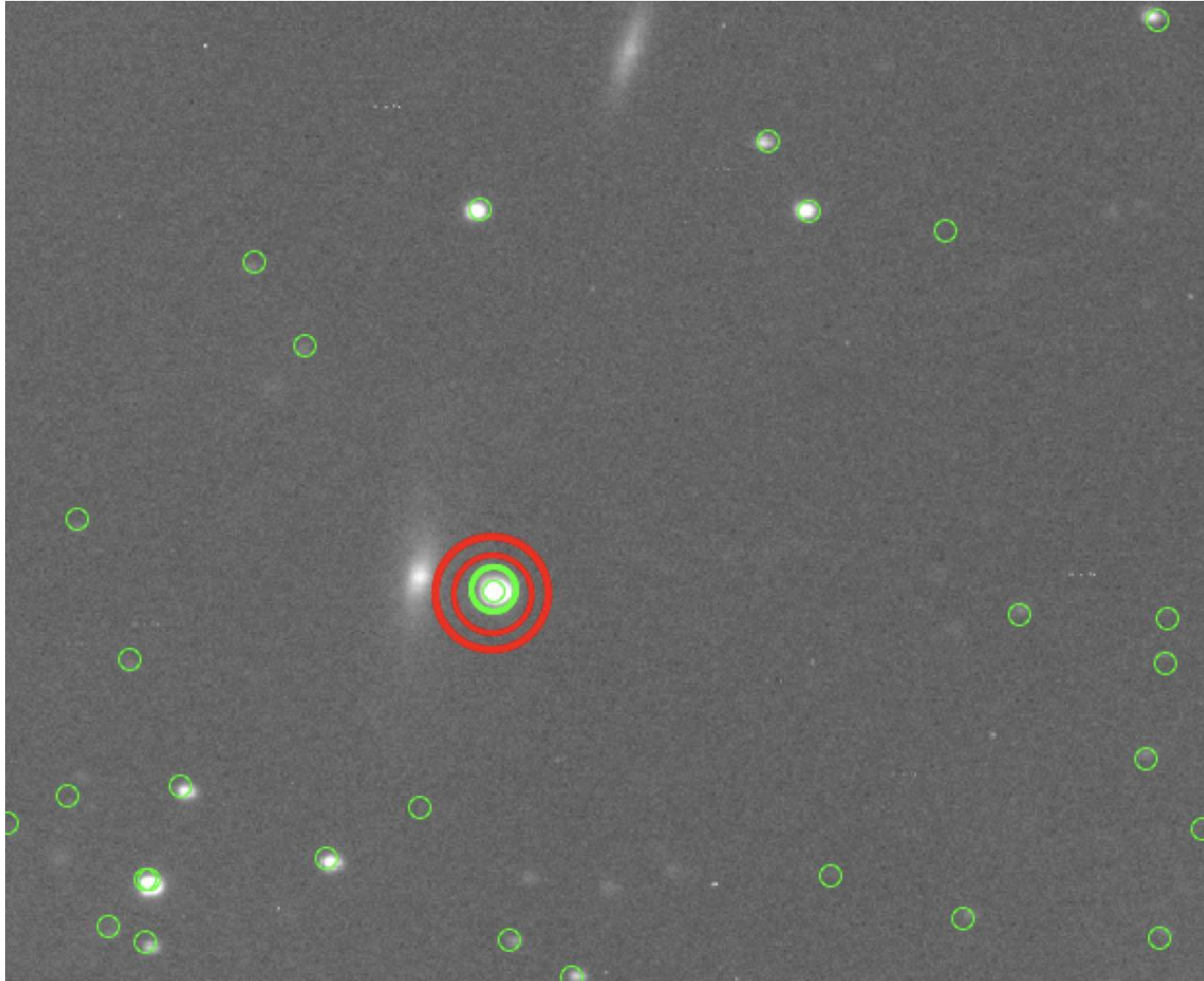


Photo from Peter Milne

Star #	Apparent magnitude	Instrumental magnitude	Offset
1	13.5	13.7	
2	14.2	14.4	
3	13.7	13.9	
4	15.5	15.7	
5	16.1	16.3	
6	14.2	14.4	
7	14.6	14.8	
SN		14.7	

Example: Calibrating Supernova Photometry

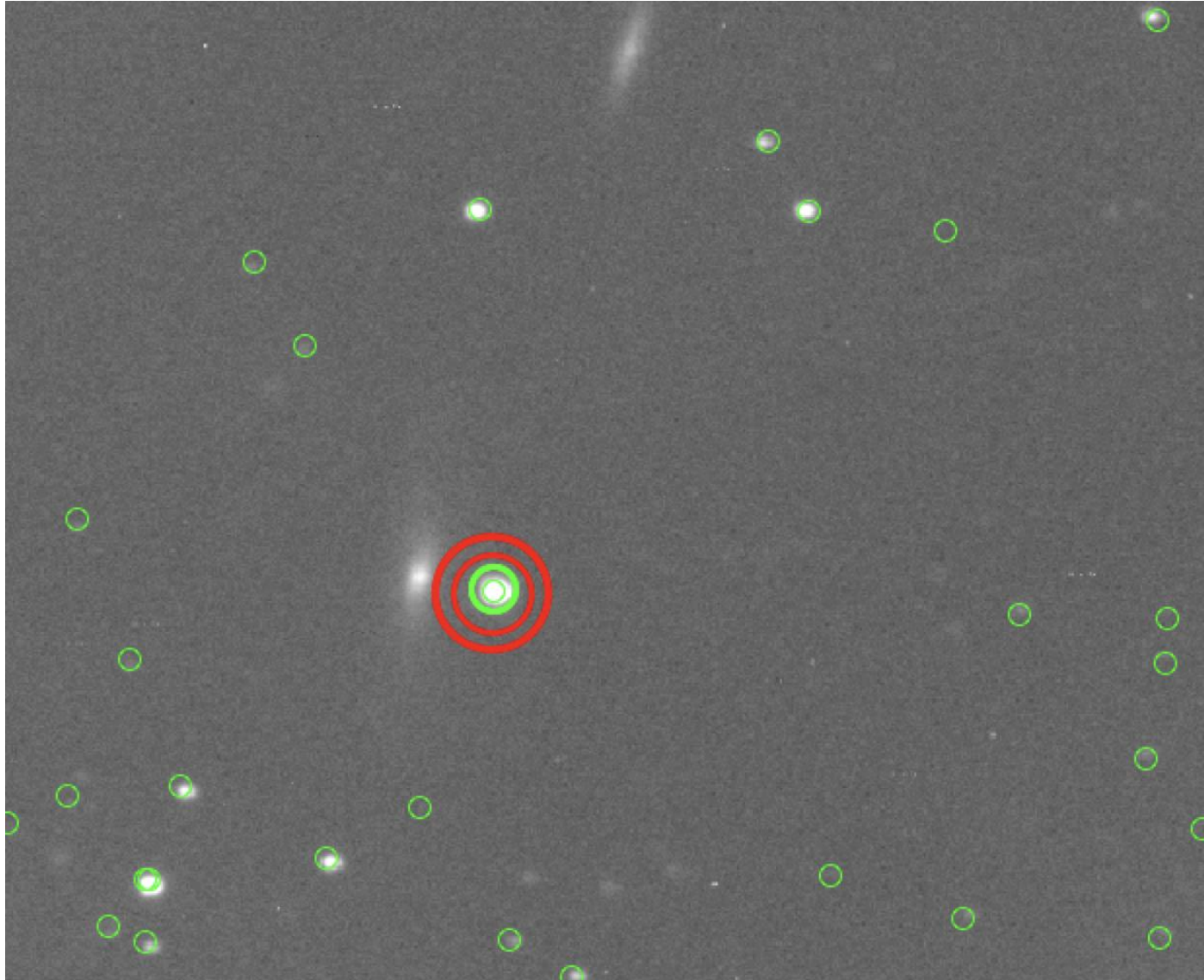


Photo from Peter Milne

Star #	Apparrent magnitude	Instrumental magnitude	Offset
1	13.5	13.7	0.2
2	14.2	14.3	0.1
3	13.7	13.9	0.2
4	15.5	15.6	0.1
5	16.1	16.4	0.3
6	14.2	14.4	0.2
7	14.6	14.9	0.3
SN	14.5	14.7	

Example: Calibrating Supernova Photometry

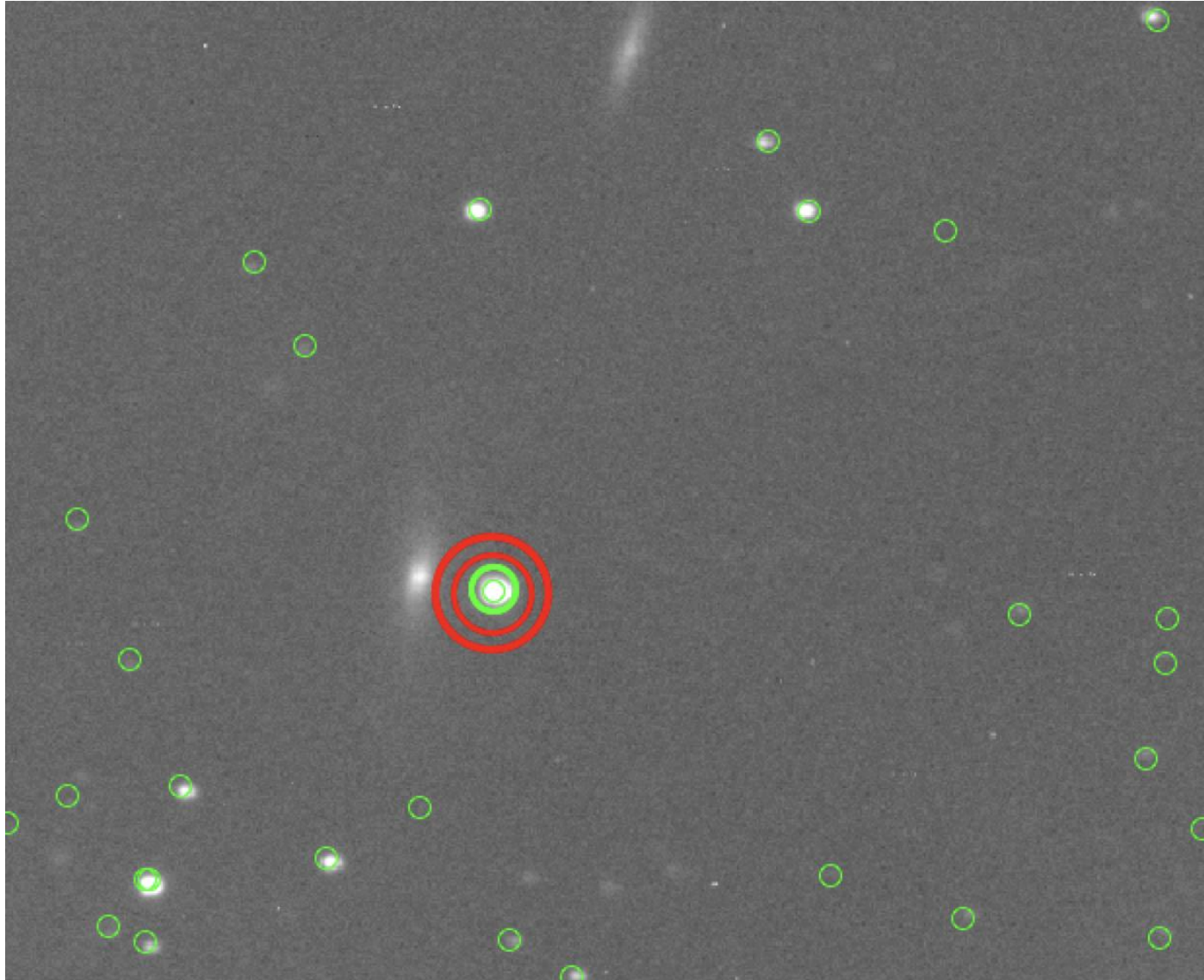


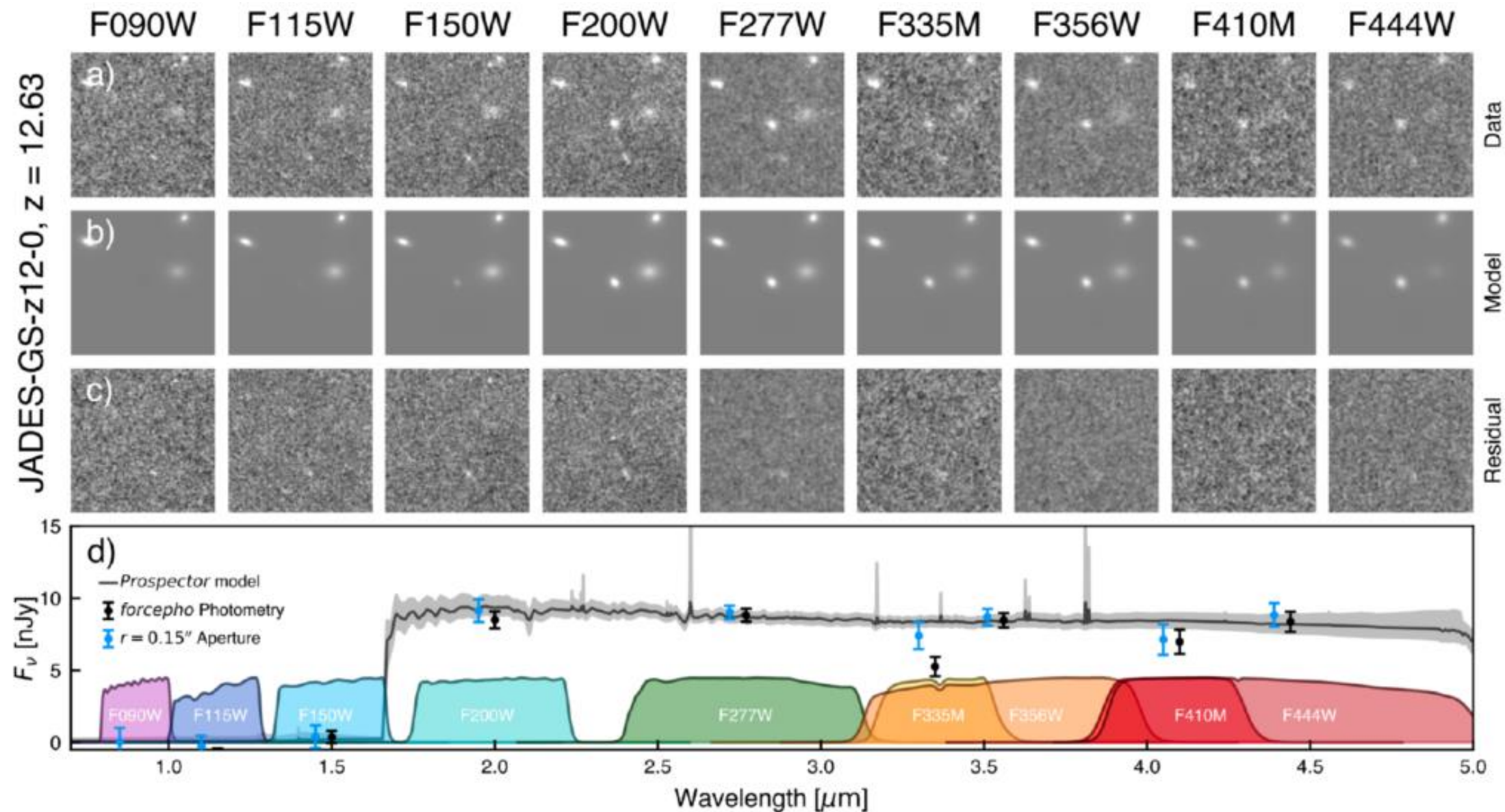
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3	13.7	13.9	0.2
4	15.5	15.6	0.1
5	16.1	16.4	0.3
6	14.2	14.4	0.2
7	14.6	14.9	0.3
SN	14.5	14.7	

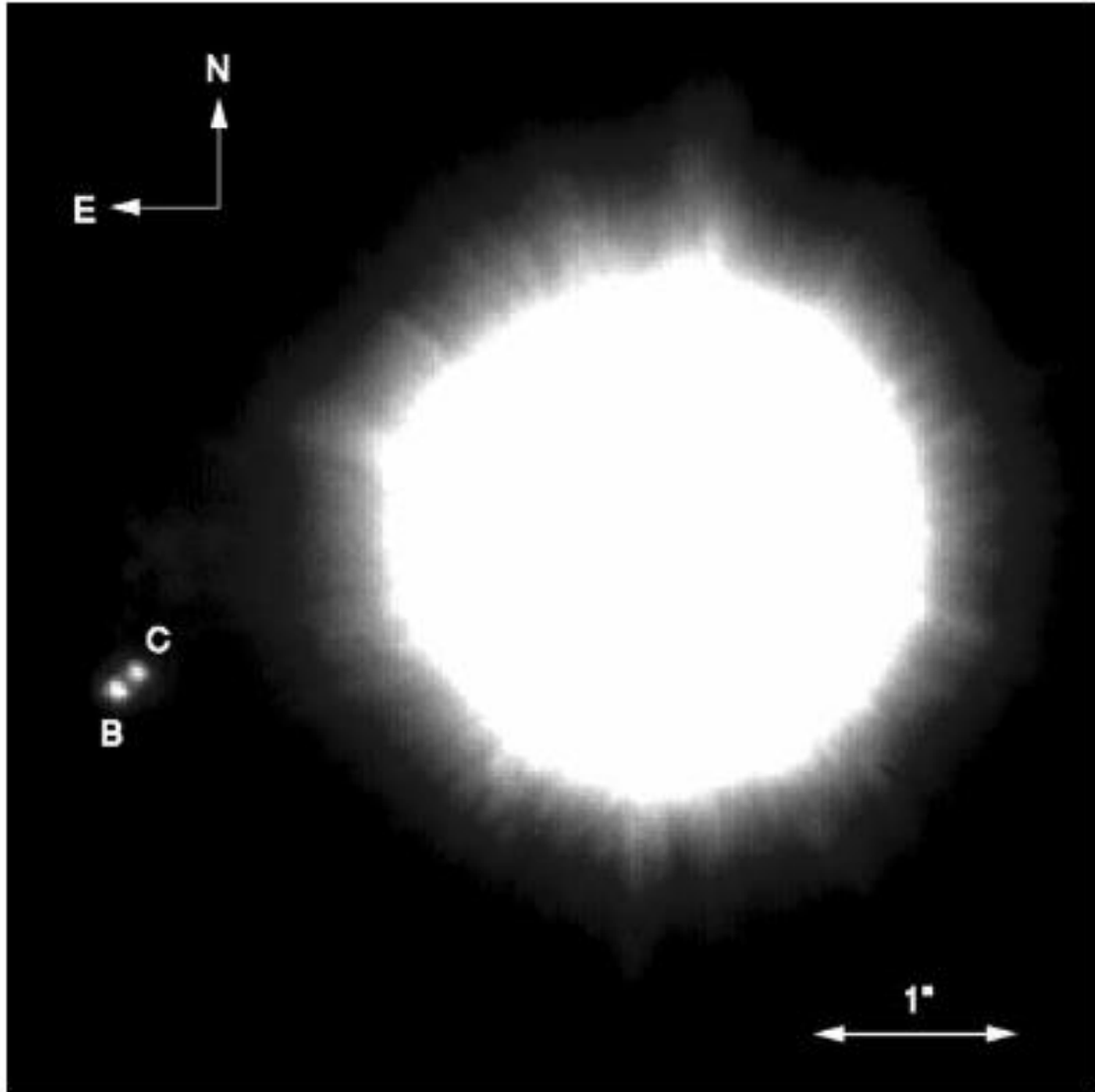
14.5 +/- 0.11

Mean offset = 0.2
STDEV = 0.1

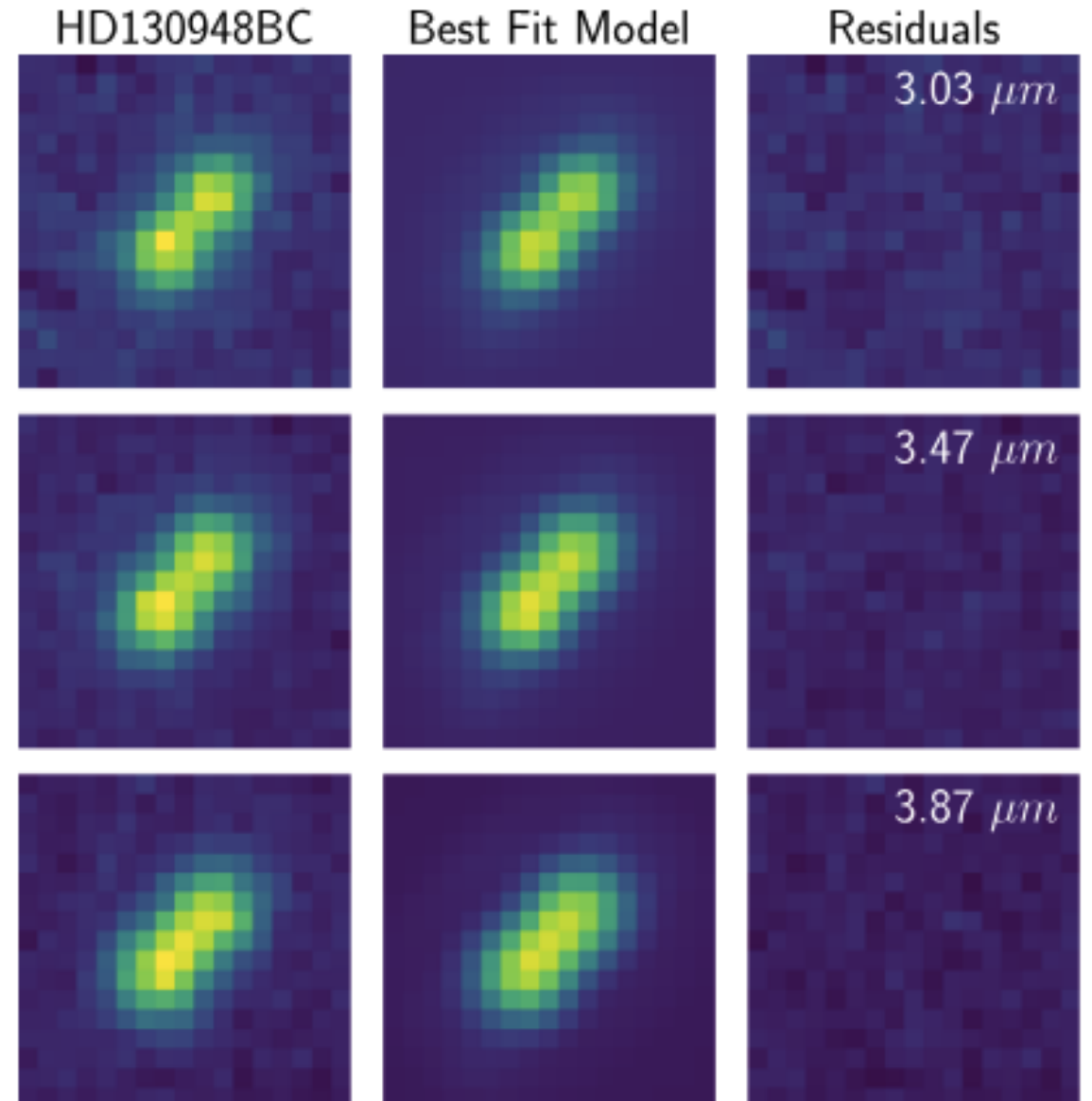
Example: Point Spread Function (PSF) Fitting



Example: PSF Fitting



Gemini North/Hokupa'a Potter+ 2002



LBT/ALES Briesemeister+ 2023